



SYSTEMS ANALYSIS AND DESIGN

IS104





INTRODUCTION TO SYSTEMS ANALYSIS AND DESIGN

INFORMATION SYSTEM COMPONENTS

TYPES OF SYSTEMS

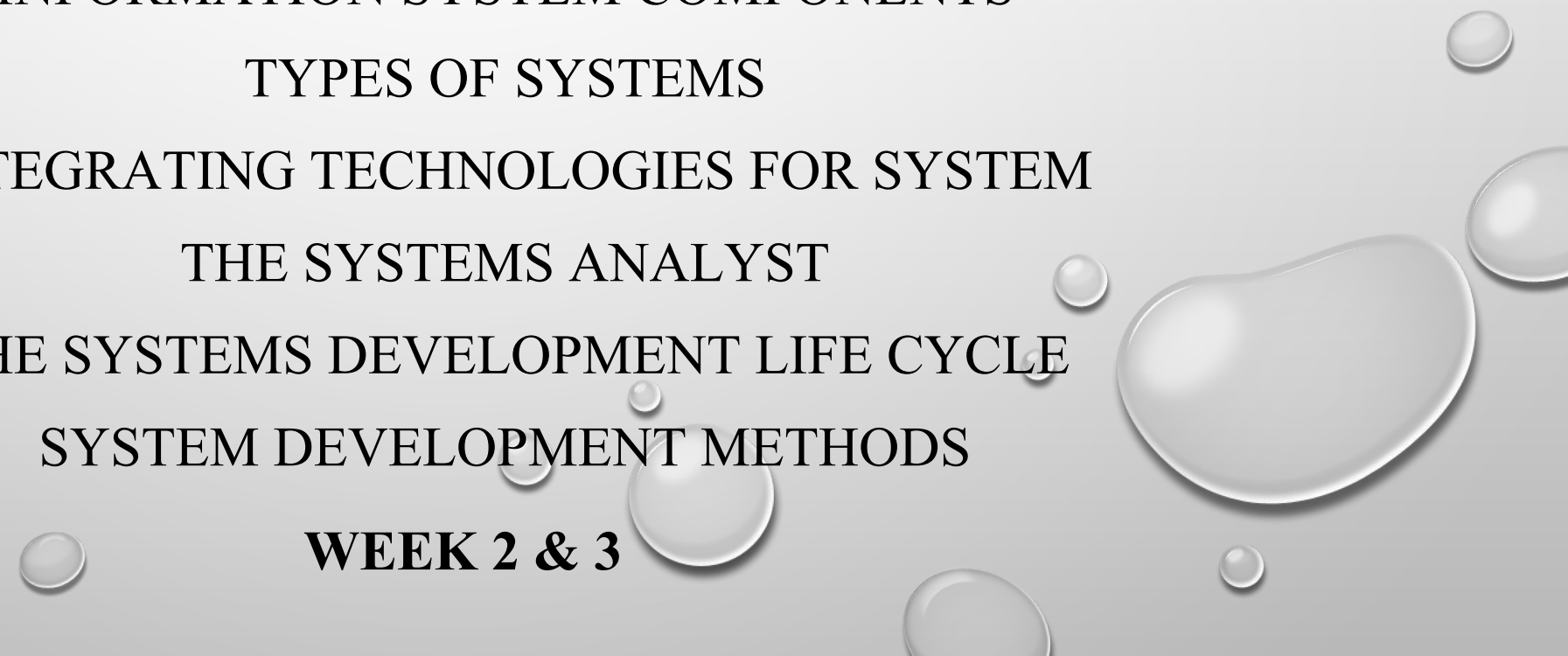
INTEGRATING TECHNOLOGIES FOR SYSTEM

THE SYSTEMS ANALYST

THE SYSTEMS DEVELOPMENT LIFE CYCLE

SYSTEM DEVELOPMENT METHODS

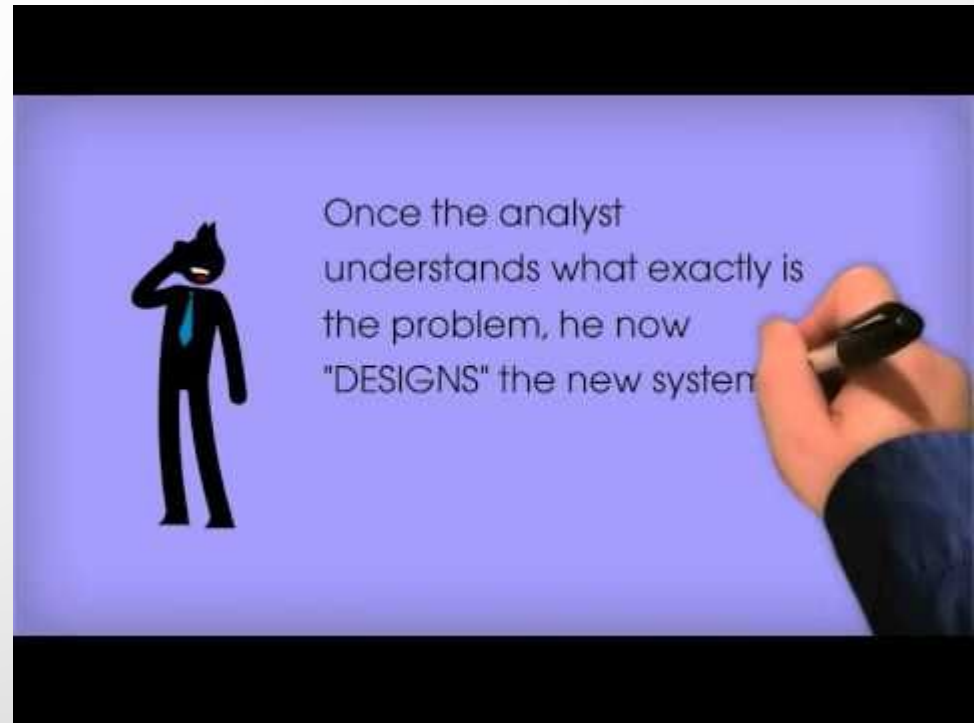
WEEK 2 & 3



Introduction to System Analysis and Design

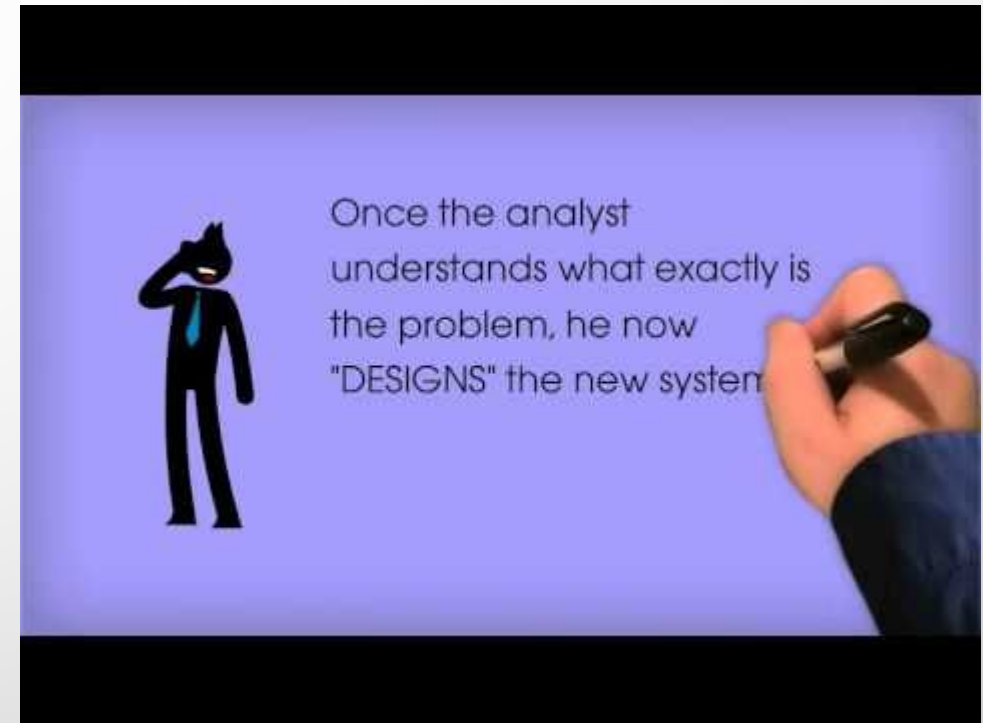
The systems development life cycle (SDLC) is the process of establishing how an information system (IS) may meet business goals, designing it, constructing it, and delivering it to users. If you've ever attended a programming class or programmed on your own, this should come as no surprise.

A team of business and systems professionals employs information systems analysis and design to construct and maintain computer-based information systems. Despite the fact that improvements in information technology provide us with new capabilities on a regular basis, the analysis and design of information systems are mostly driven by organizational needs.



Introduction to System Analysis and Design

Benefits increases as a result of adding value to the organization's products and services during the design, production, and support processes. As a result, information system analysis and design are based on your knowledge of the organization's goals, structure, and procedures, as well as your awareness of how to use information technology to your benefit.



Information system

System – interrelated components functioning together to achieve an outcome.

Information system (IS) - is an arrangement of people, data, processes, and information technology that interact to collect, process, store, and provide as output the information needed to support an organization.

Information technology - is a contemporary term that describes the combination of computer technology (hardware and software) with telecommunications technology (data, image, and voice networks).



Components of Information Systems

- ✓ Resources of people: (end users and IS specialists, system analyst, programmers, data administrators etc.).
- ✓ Hardware: (Physical computer equipments and associate device, machines and media).
- ✓ Software: (programs and procedures).
- ✓ Data: (data and knowledge bases), and
- ✓ Networks: (communications media and network support).



Types of Information Systems

Transaction Processing Systems (TPS)

These systems have been designed to collect, process and store transactions that occur in the day-to-day operations of a company.

The system can also be used to cancel or modify transactions done in the past if the need arises. One property of this system that enables them to work effectively is the ability to accurately record multiple transactions even if the different transactions take place simultaneously.

They are built to be able to handle large volumes of transactions. Examples include stock control systems, payroll systems, order processing systems etc.



Types of Information Systems

Office Automation Systems

Office automation systems (OAS) support data workers, who do not usually create new knowledge but rather analyze information to transform data or manipulate it in some way before sharing it with, or formally disseminating it throughout, the organization and, sometimes, beyond. Familiar aspects of OAS include word processing, spreadsheets, desktop publishing, electronic scheduling, and communication through voice mail, email (electronic mail), and teleconferencing.



Types of Information Systems

Knowledge Work Systems

Knowledge work systems (KWS) support professional workers such as scientists, engineers, and doctors by aiding them in their efforts to create new knowledge (often in teams) and by allowing them to contribute it to their organization or to society at large.



6 Stages of Knowledge Management

Types of Information Systems

Management Information Systems

Management information systems are regarded to be a subset of the overall internal controls procedures in a business, which cover the application of people, documents, technologies, and procedures used by management accountants to solve business problems such as costing a product, service or a business-wide strategy.



Components of MIS

Marketing Research System (MRS)

Marketing research can be seen as the systematic and objective search for and analysis of data and information relevant to the identification and solution of any problem in the field of marketing.



Components of MIS

Marketing Intelligence System (MIS)

The process of acquiring and analyzing information in order to understand the market (both existing and potential customers); to determine the current and future needs and preferences, attitudes and behavior of the market; and to assess changes in the business environment that may affect the size and nature of the market in the future.

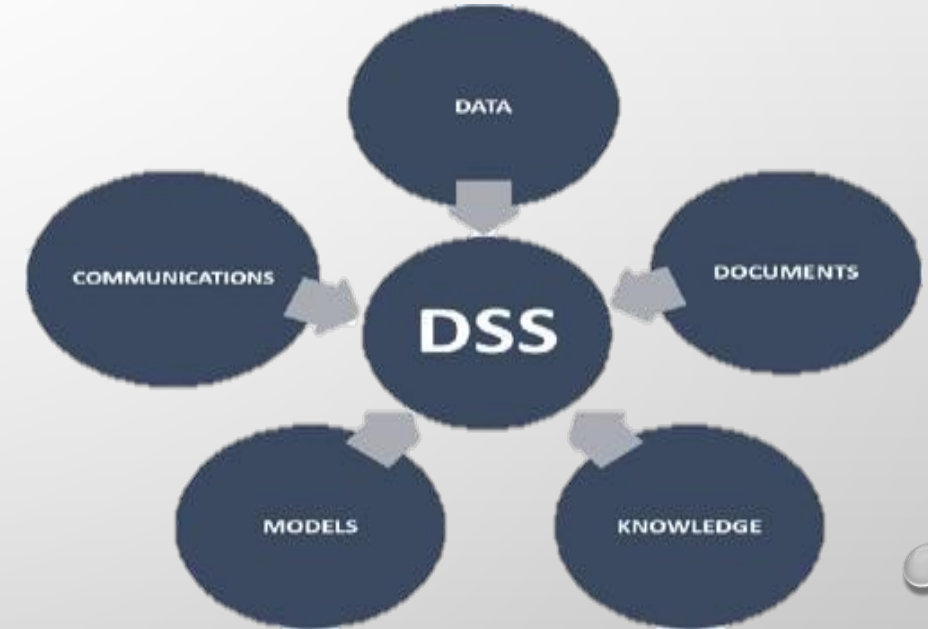


Components of MIS

Decision Support System(DSS)

A decision support system (DSS) is a computer-based information system that supports business or organizational decision-making activities.

DSSs serve the management, operations, and planning levels of an organization and help to make decisions, which may be rapidly changing and not easily specified in advance.



Components of MIS

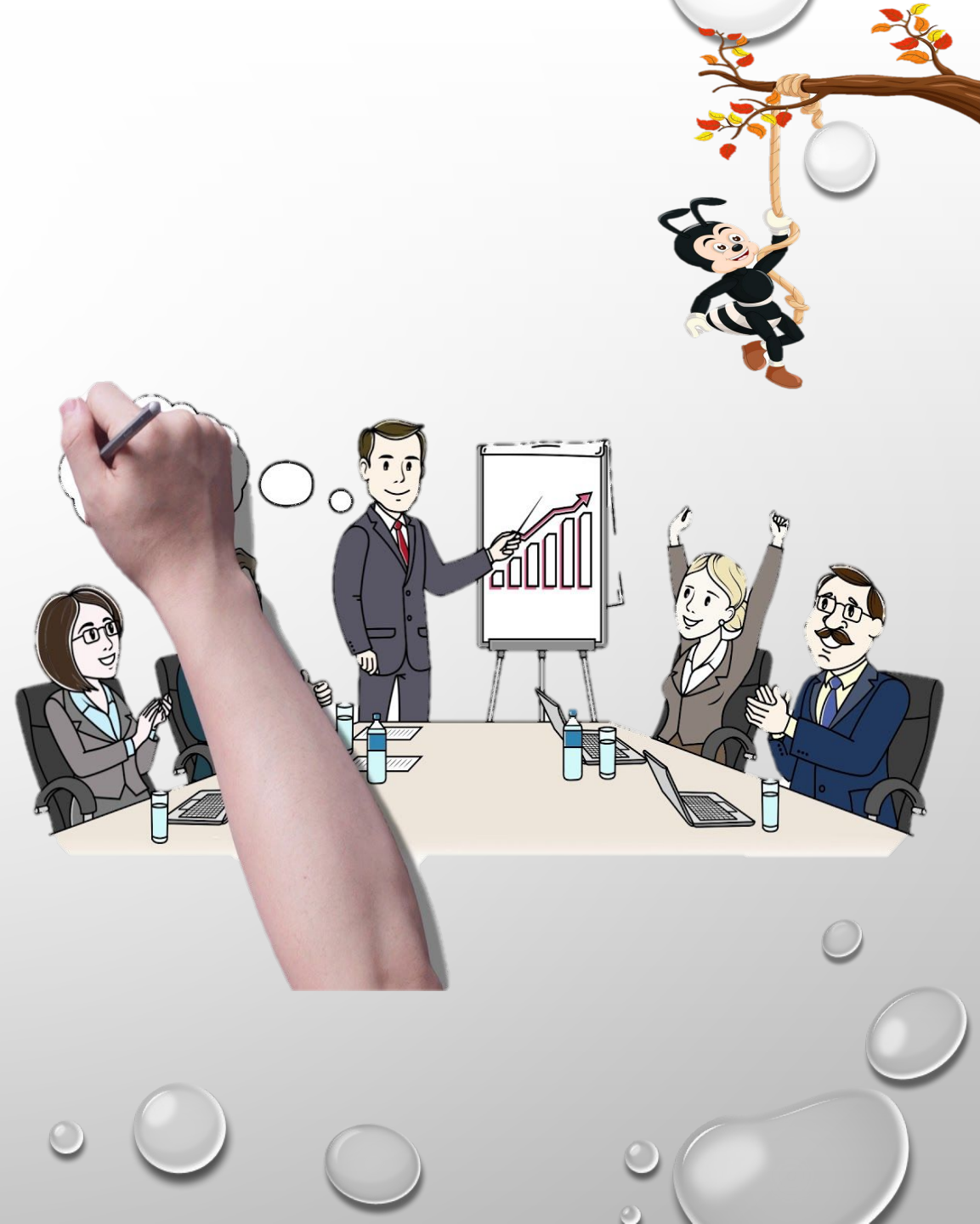
Executive Support Systems

An executive support system (ESS) – an extension of MIS – is a computer-based information system that helps in decision-making at the top level of an organization. The decisions taken with the help of an executive support system are non-routine decisions that affect the entire organization and, thus, require judgment and sight.

As compared to DSSs, ESSs offer more general computing capabilities, better telecommunications, and efficient display options. They use advanced graphics software to display critical information in the form of charts or graphs that help senior executives to solve a wide range of problems.

To make effective decisions, they use summarized internal data from MIS and DSS as well as data from external sources about events like new tax laws, new competitors, etc. They filter, compress, and track data of high importance and make it available to the strategic-level managers.

Executive support systems help to monitor performance, track activities of competitors, identify opportunities, and forecast trends.



Components of MIS

Artificial Intelligence and Expert Systems

Artificial intelligence (AI) can be considered the overarching field for expert systems. The general thrust of AI has been to develop machines that behave intelligently.

Two avenues of AI research are

- (1) understanding natural language
- (2) analyzing the ability to reason through a problem to its logical conclusion.

Expert systems use the approaches of AI reasoning to solve the problems put to them by business (and other) users.



Components of MIS

Enterprise Systems

Enterprise systems, also called enterprise resource planning (ERP) systems, are designed to perform this integration. Instituting ERP requires enormous commitment and organizational change. Often systems analysts serve as consultants to ERP endeavors that use proprietary software. Popular ERP software includes that from SAP and Oracle. Some of these packages are targeted toward moving enterprises onto the Web. Typically, analysts as well as some users require vendor training, support, and maintenance to be able to properly design, install, maintain, update, and use a particular ERP package.



Components of MIS

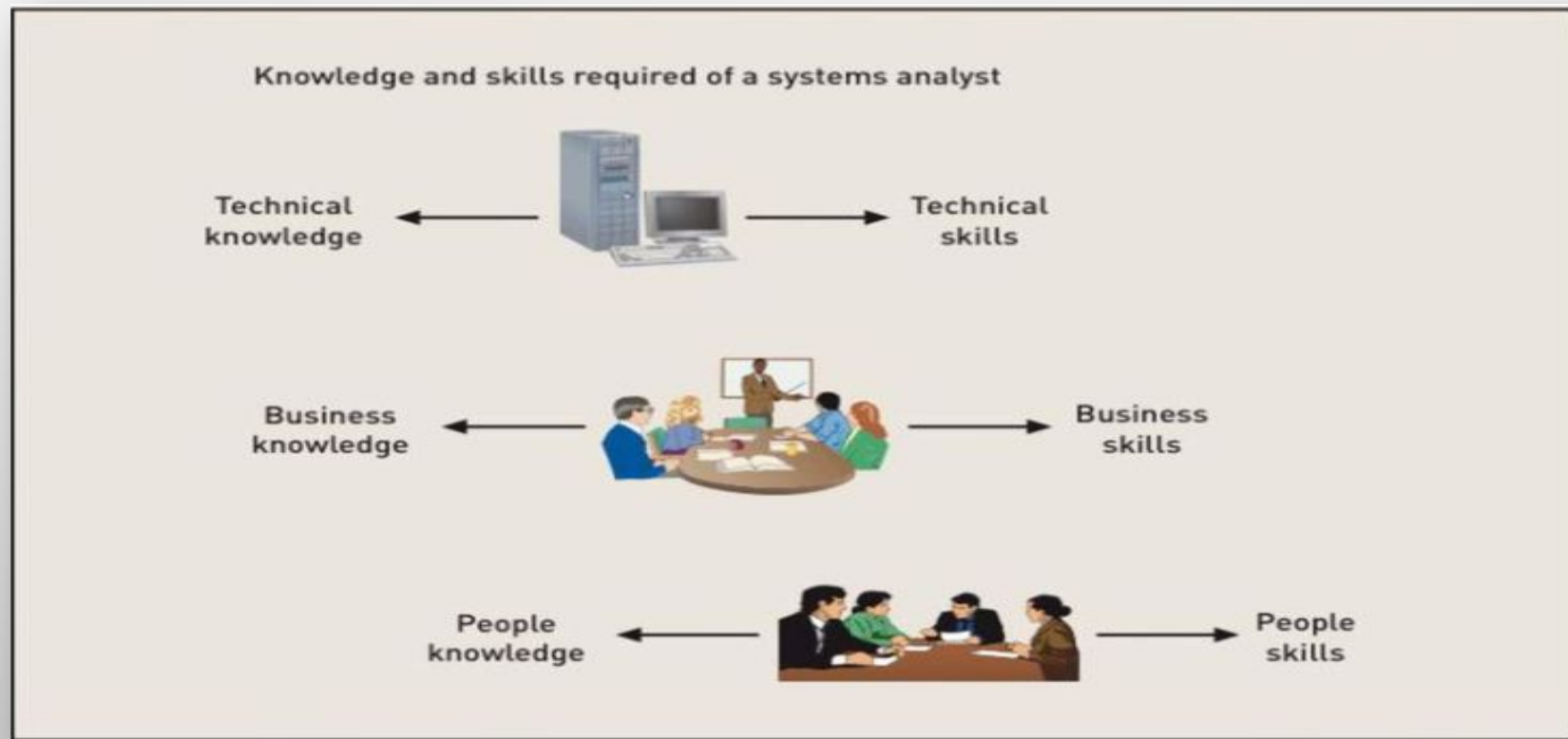
Systems Analyst

An IT professional involved in the development of a computerized solution to a business problem Requires extensive technical, business and people knowledge, communication, business and technical skills Is fundamentally curious to explore how things are done with a desire to make them work better.



Components of MIS

Required Skills of System Analysts



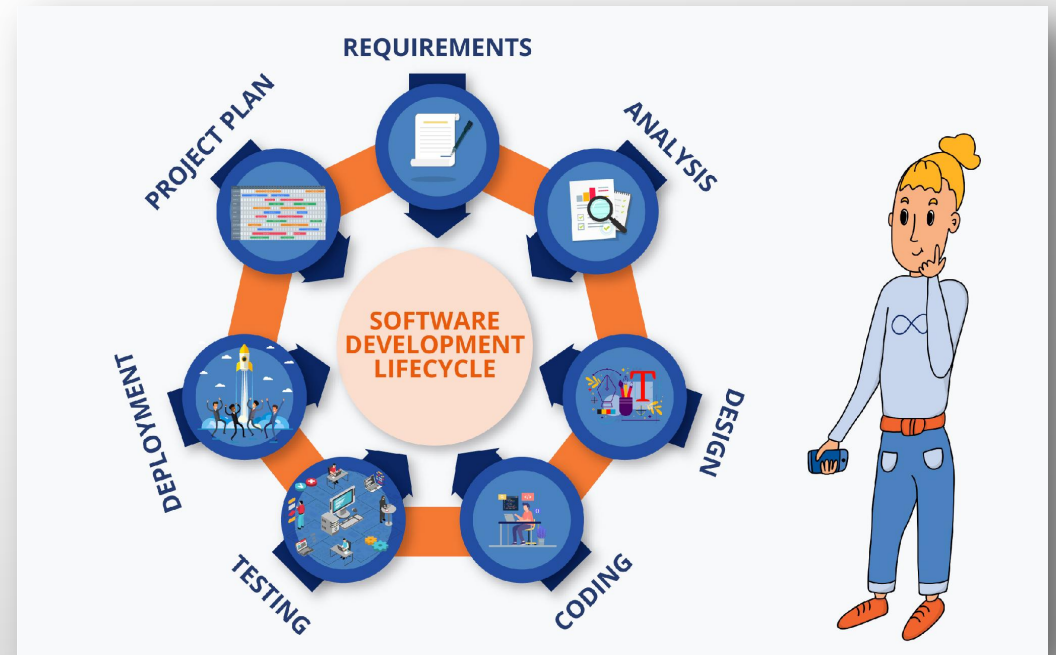


The Systems Development Life Cycle (SDLC)

Refers to stages in the conception, design, creation, and implementation of an information system.

Systems analysis consists of those activities that enable a person to understand and specify what the new system should accomplish.

Systems design consists of those activities that enable a person to describe in detail how the information system will actually be implemented to provide the needed solution. In other words, systems design describes *how* the system will actually work. It specifies in detail all the components of the solution system and how they work together.

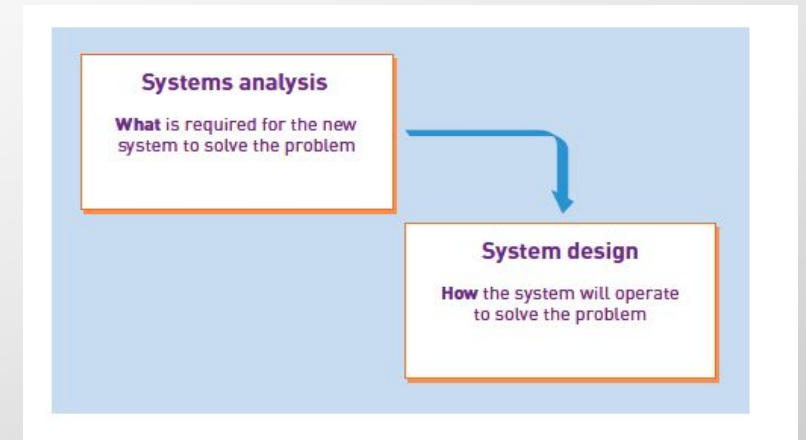
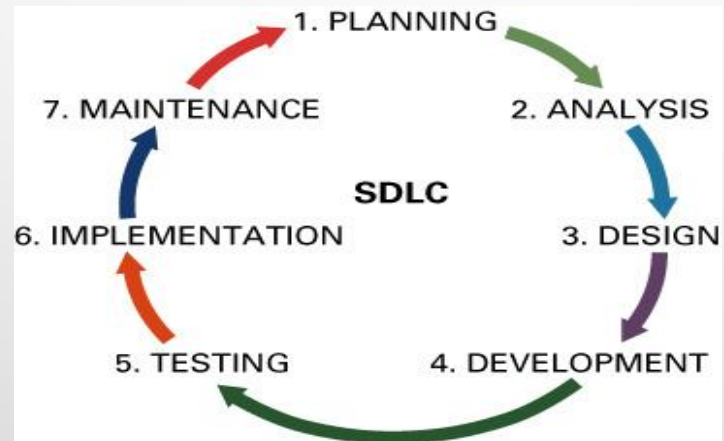




The Systems Development Life Cycle (SDLC)

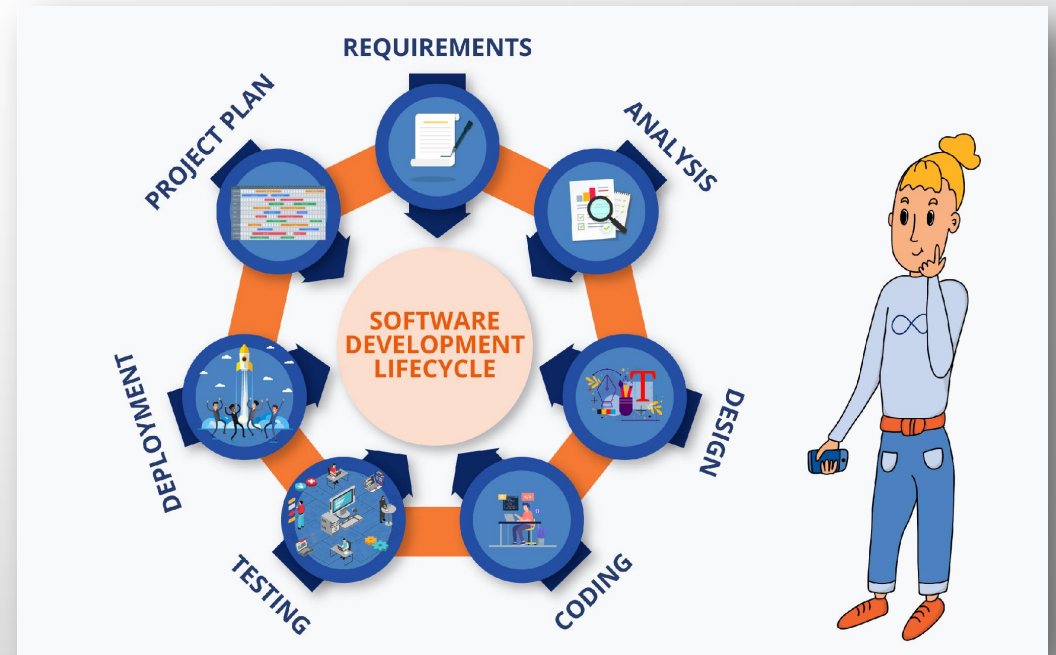
Phases of SDLC

1. Planning
2. Analysis
3. Design
4. Development
5. Testing
6. Implementation
7. Maintenance



The Systems Development Life Cycle (SDLC)

Software development life cycle (SDLC) is a framework defining tasks performed at each step in the software development process. It consists of a detailed plan describing how to develop, maintain, replace and alter or enhance specific software.





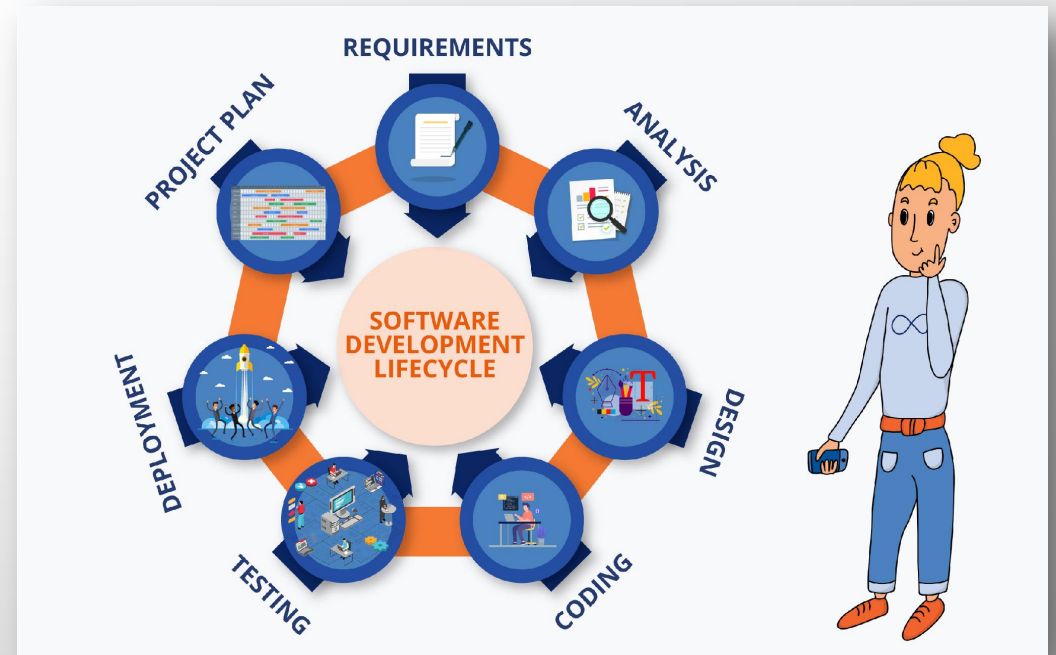
The Systems Development Life Cycle (SDLC)

Phase 1: Planning

Involves determining a solid plan for developing your information system

Three primary planning activities:

- Define the system to be developed
- Set the project scope
- Develop the project plan including tasks, resources, and timeframes



The Systems Development Life Cycle (SDLC)

Phase 2: Analysis

Involves end users and IT specialists working together to gather, understand, and document the business requirements for the proposed system

Two primary analysis activities:

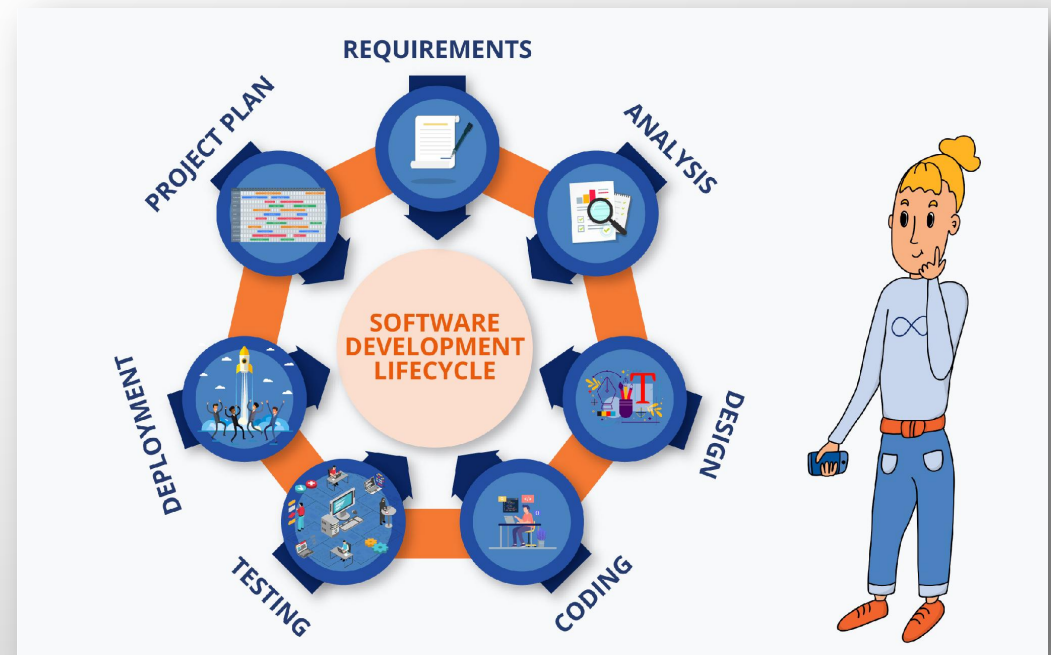
1. Gather the business requirements

Business requirements - the detailed set of knowledge worker requests that the system must meet in order to be successful.

Joint application development (JAD) - knowledge workers and IT specialists meet, sometimes for several days, to define or review the business requirements for the system.

2. Prioritize the requirements

Requirements definition document – prioritizes the business requirements and places them in a formal comprehensive document



The Systems Development Life Cycle (SDLC)

Phase 3: Design

Build a technical blueprint of how the proposed system will work

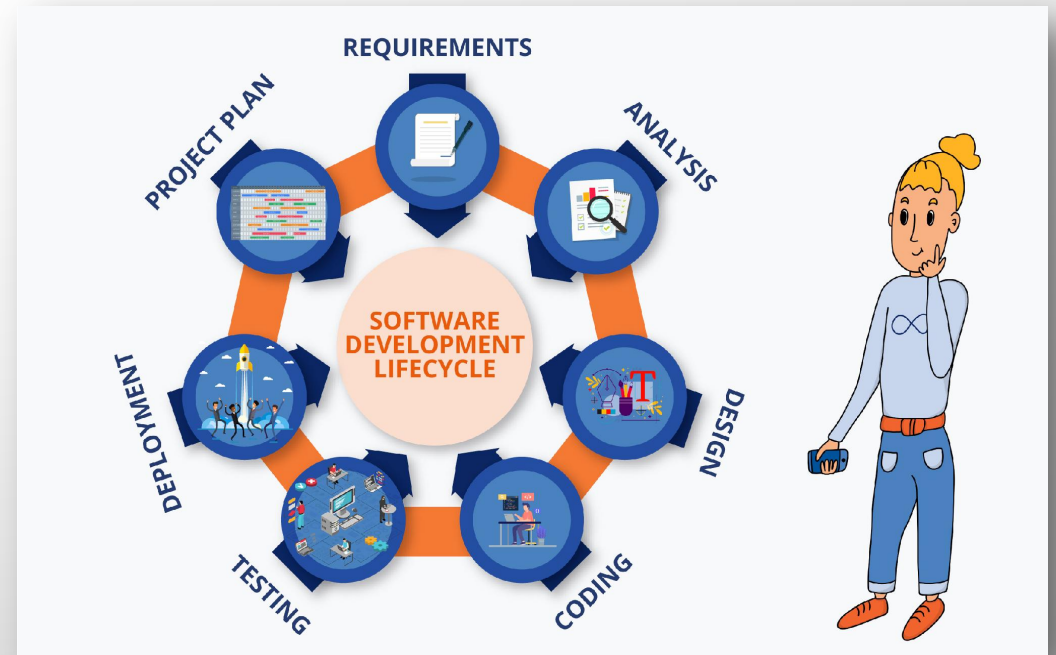
Two primary design activities:

1.Design the technical architecture

- Technical architecture - defines the hardware, software, and telecommunications equipment required to run the system

2.Design system models

- Modeling - the activity of drawing a graphical representation of a design
- Graphical user interface (GUI) - the interface to an information system
- GUI screen design - the ability to model the information system screens for an entire system



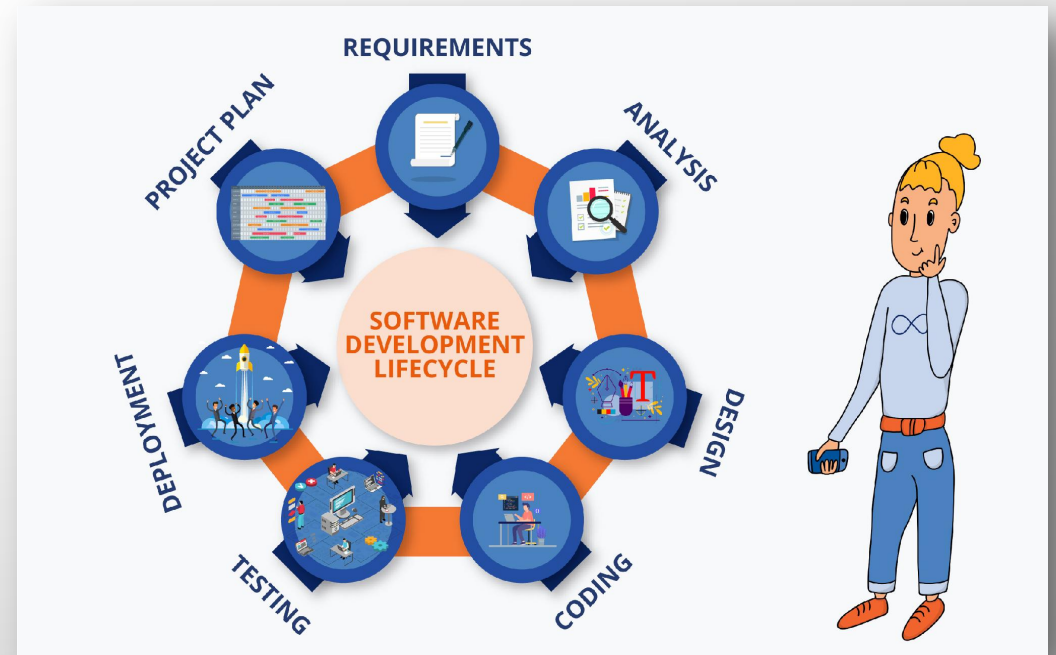
The Systems Development Life Cycle (SDLC)

Phase 4: Development

Take all of your detailed design documents from the design phase and transform them into an actual system.

Two primary development activities:

3. Build the technical architecture
4. Build the database and programs



The Systems Development Life Cycle (SDLC)

Phase 5: Testing

Verifies that the system works and meets all of the business requirements defined in the analysis phase.

Two primary testing activities:

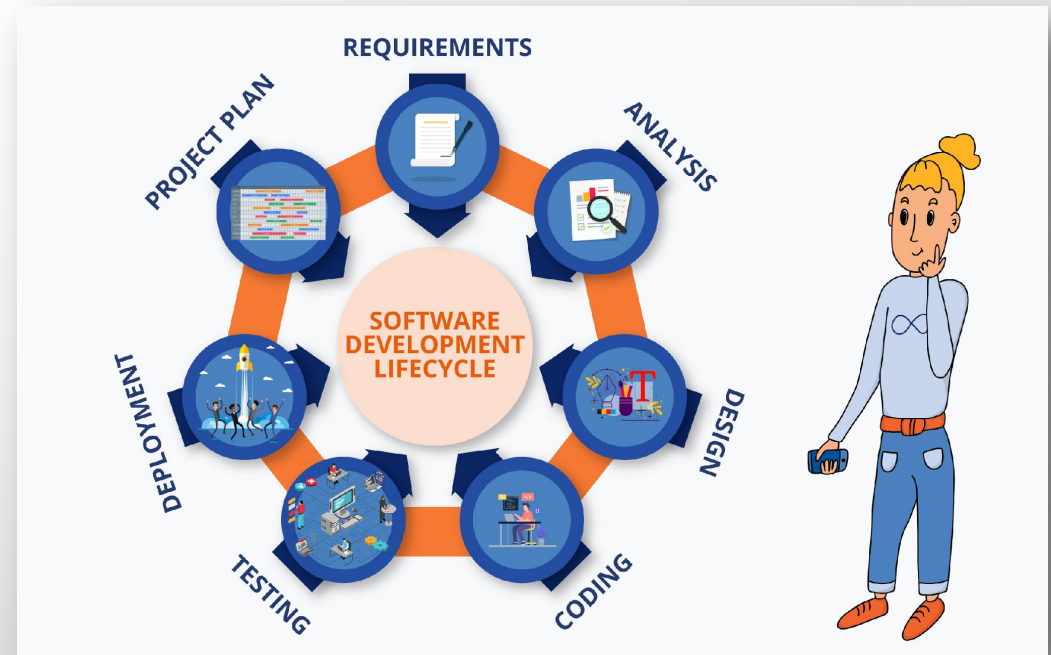
1. Write the test conditions

- Test conditions - the detailed steps the system must perform along with the expected results of each step

2. Perform the testing of the system

- a. Unit testing – tests individual units of code
- b. System testing – verifies that the units of code function correctly when integrated
- c. Integration testing – verifies that separate systems work together
- d. User acceptance testing (UAT) – determines if the system

satisfies the business requirements



The Systems Development Life Cycle (SDLC)

Phase 6: Implementation

Distribute the system to all of the knowledge workers and they begin using the system to perform their everyday jobs.

Two primary implementation activities

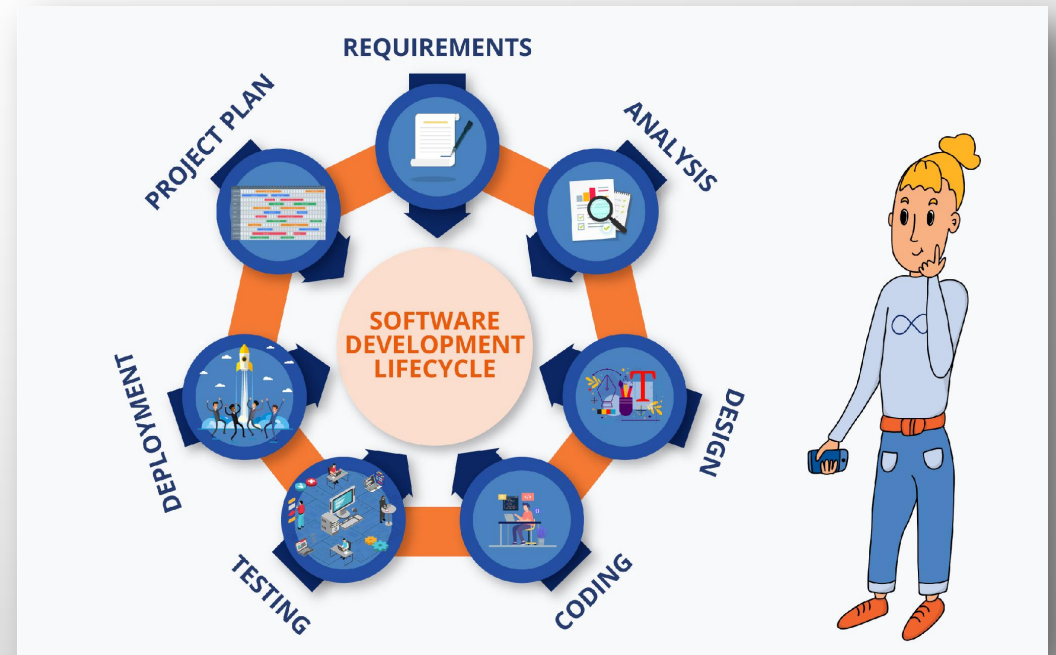
1. Write detailed user documentation

User documentation - highlights how to use the system

2. Provide training for the system users:

Online training - runs over the Internet or off a CD-ROM

Workshop training - is held in a classroom environment and lead by an instructor



The Systems Development Life Cycle (SDLC)

Phase 7: Maintenance

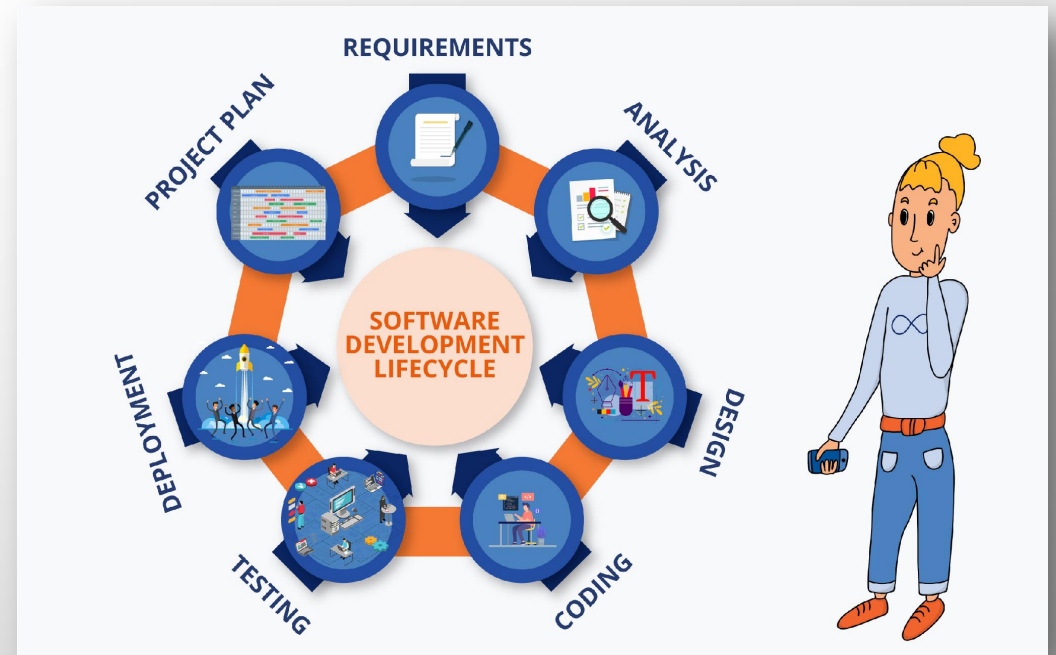
Monitor and support the new system to ensure it continues to meet the business goals.

Two primary maintenance activities:

1. Build a help desk to support the system users

Help desk - a group of people who responds to knowledge workers' questions

2. Provide an environment to support system changes



SYSTEMS DEVELOPMENT METHODOLOGIES



Software development methodology involves different processes for developing the software project and follows a development life cycle. The methodology is selected based on the aims and objectives of the project.

Software development methodology is an approach in software engineering that provides a framework for defining the structure, plan, and control of different processes while developing an information system (Vick, 1984).

There are six more general categories/approaches, which are developed into several specific methodologies:

1. Waterfall - Linear framework type.
2. Prototyping - Iterative framework type
3. Incremental- Combination of linear and iterative framework type
4. Spiral - Combination of linear and iterative framework type
5. Rapid Application Development (RAD) - Iterative Framework Type.
6. Agile Development- Based on iterative development emphasis on teams



SYSTEMS DEVELOPMENT METHODOLOGIES

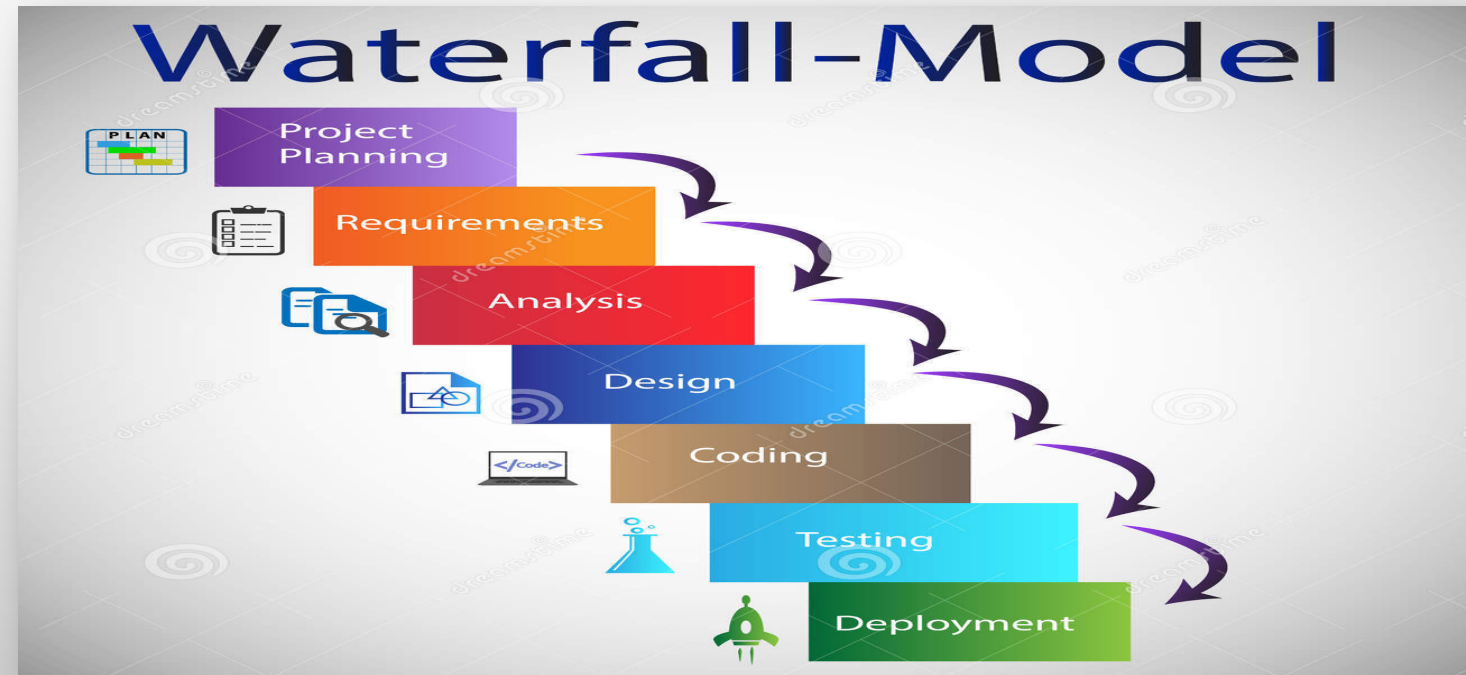


1. Waterfall Methodology

A sequential, activity-based process in which each phase in the SDLC is performed sequentially from planning through implementation.

The waterfall methodology is composed of seven non-overlapping stages:

1. Planning
2. Analysis
3. Design
4. Development
5. Testing
6. Implementation
7. Maintenance





WATERFALL METHODOLOGY STAGES

1. Planning

Potential requirements, deadlines and guidelines for the project are analyzed and placed into a functional specification. This stage handles the defining and planning of the project without mentioning specific processes.

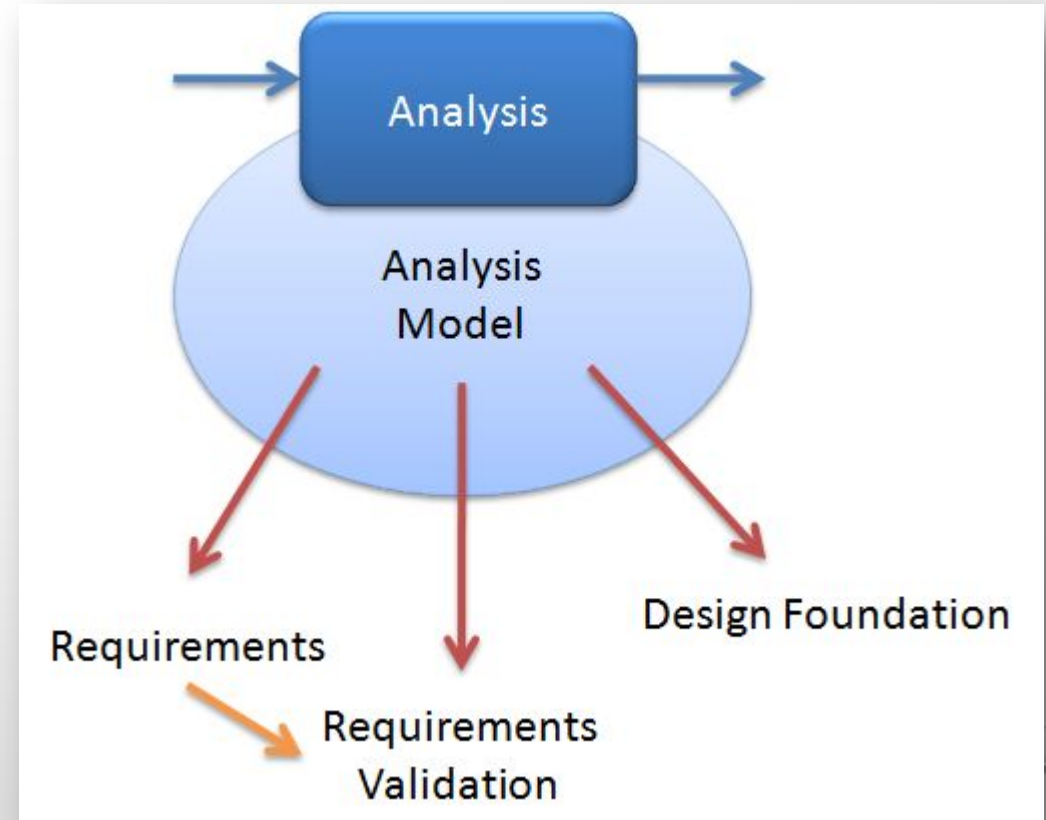




WATERFALL METHODOLOGY STAGES

2. Analysis

The system specifications are analyzed to generate product models and business logic that will guide production. This is also when financial and technical resources are audited for feasibility.

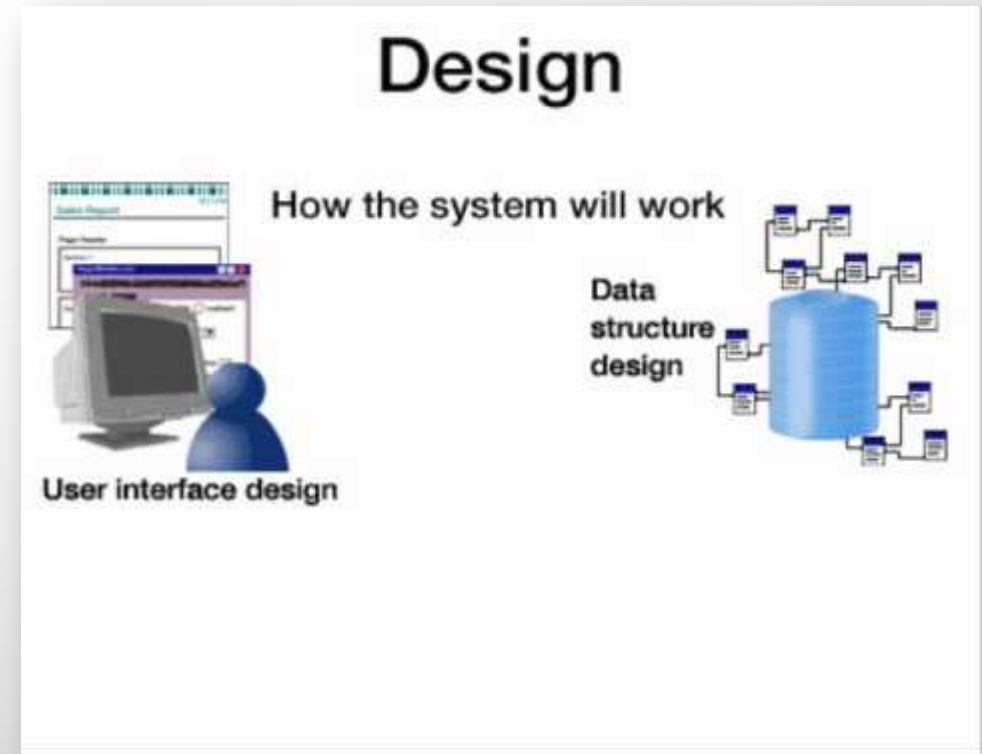




WATERFALL METHODOLOGY STAGES

3. Design

The system specifications are analyzed to generate product models and business logic that will guide production. This is also when financial and technical resources are audited for feasibility.





WATERFALL METHODOLOGY STAGES

4. Development

The source code is developed using the models, logic and requirements designated in the prior stages. Typically, the system is designed in smaller components, or units, before being implemented together.

Sitefinity development phases

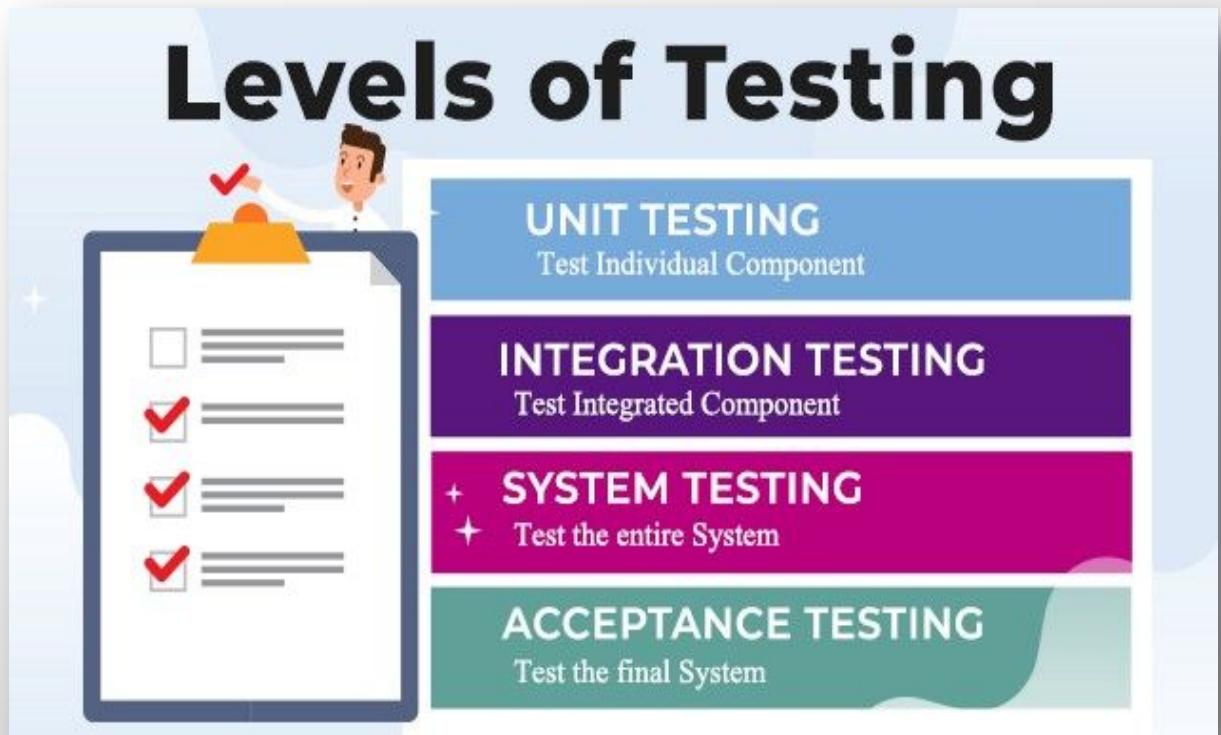




WATERFALL METHODOLOGY STAGES

5. Testing

This is when quality assurance, unit, system and beta tests take place to report issues that may need to be resolved. This may cause a forced repeat of the coding stage for debugging. If the system passes the tests, the waterfall continues forward.



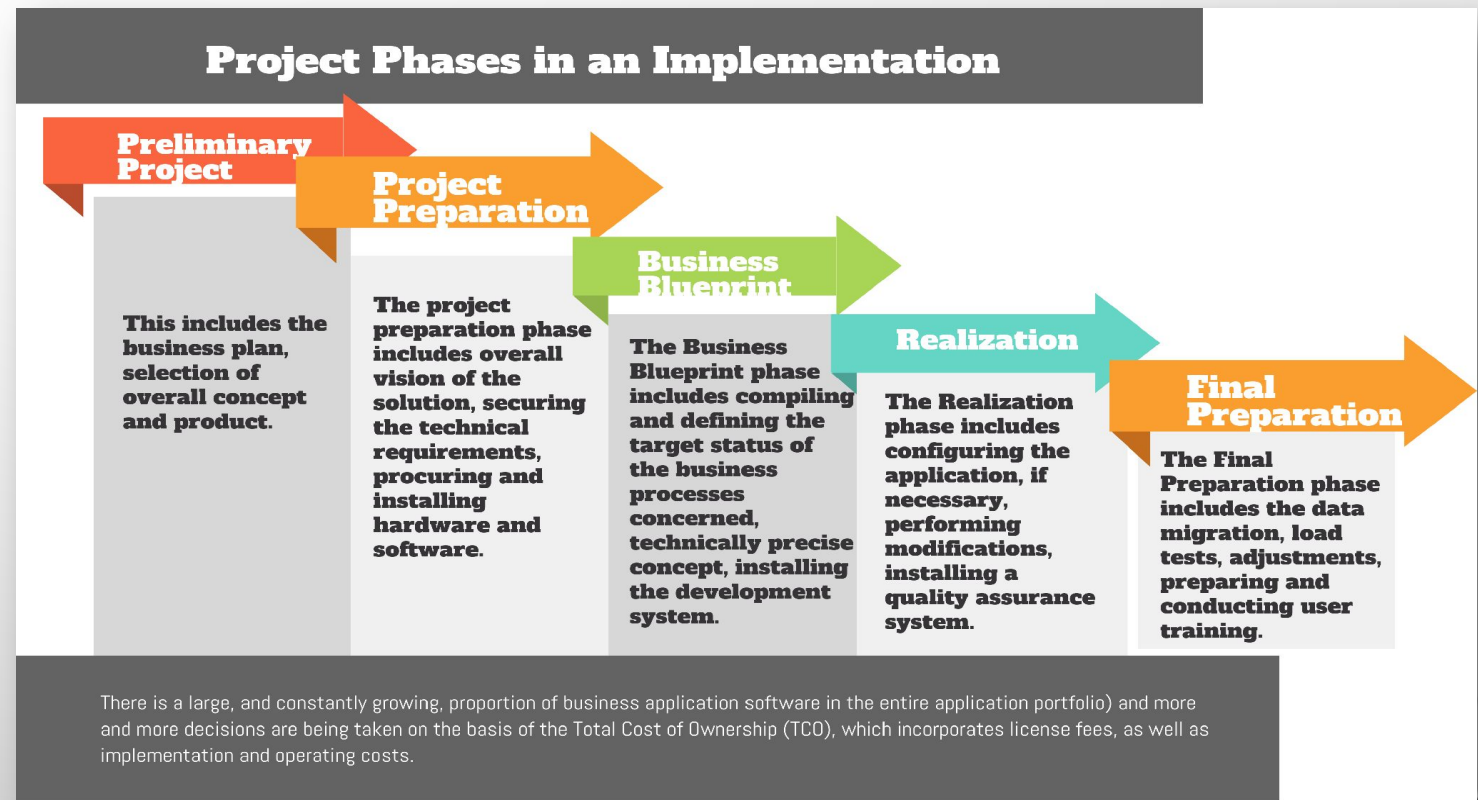


WATERFALL METHODOLOGY

STAGES

6. Implementation

The product or application is deemed fully functional and is deployed to a live environment.

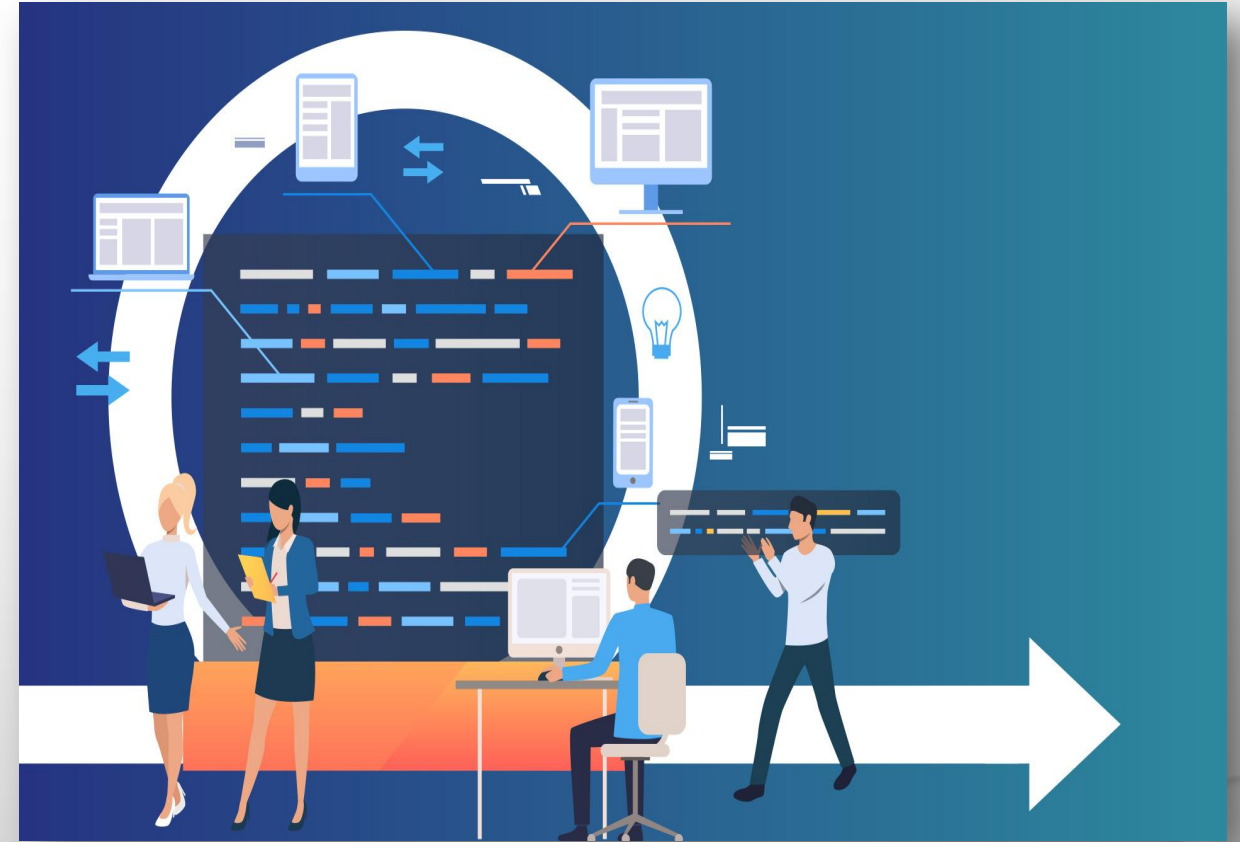




WATERFALL METHODOLOGY STAGES

7. Maintenance

Corrective, adaptive and perfective maintenance is carried out indefinitely to improve, update and enhance the final product. This could include releasing patch updates or releasing new versions.





ADVANTAGES OF WATERFALL

- ✓ Simple method and easy to use
- ✓ Phases are clear
- ✓ Suitable for smaller projects
- ✓ Easy to manage



DISADVANTAGES OF WATERFALL

- ✓ Does not allow much revision
- ✓ Not suitable for complex projects
- ✓ Risk and uncertainty are high
- ✓ Does not include a feedback path



SYSTEMS DEVELOPMENT METHODOLOGIES



2. PROTOTYPING

The process of building a model that demonstrates the features of a proposed product, service, or system.

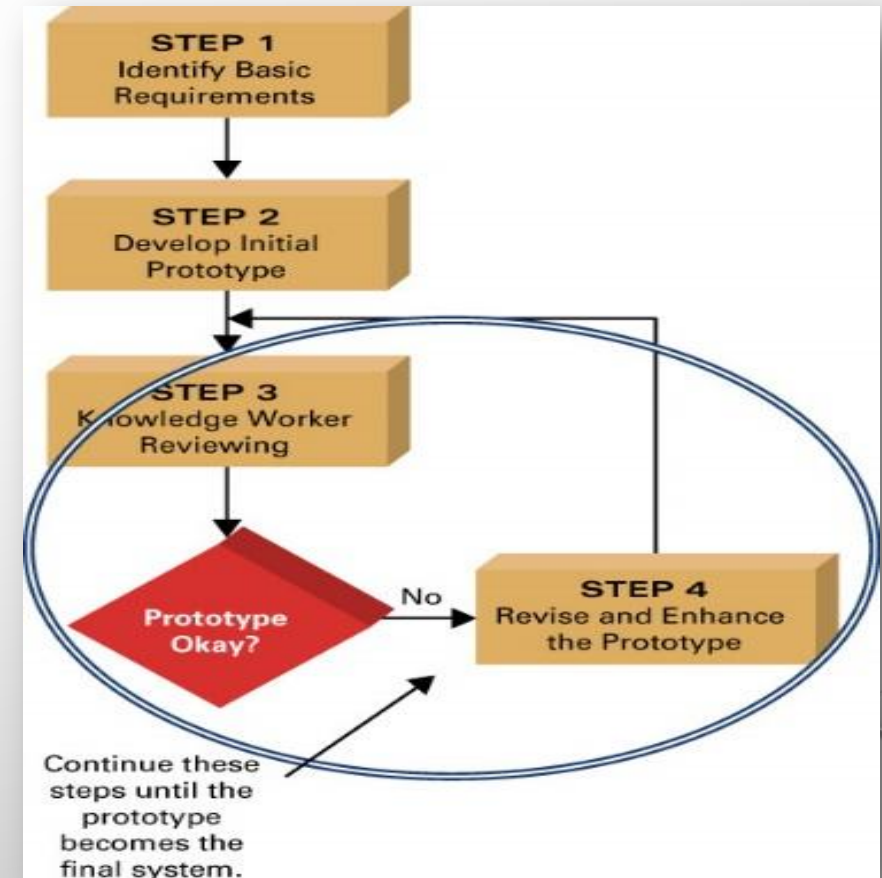
Proof-of-concept prototype - used to prove the technical feasibility of a proposed system

Selling prototype - used to convince people of the worth of a proposed system

The Prototyping Process

The prototyping process involves four steps:

1. Identify basic requirements
2. Develop initial prototype
3. User review
4. Revise and enhance the prototype

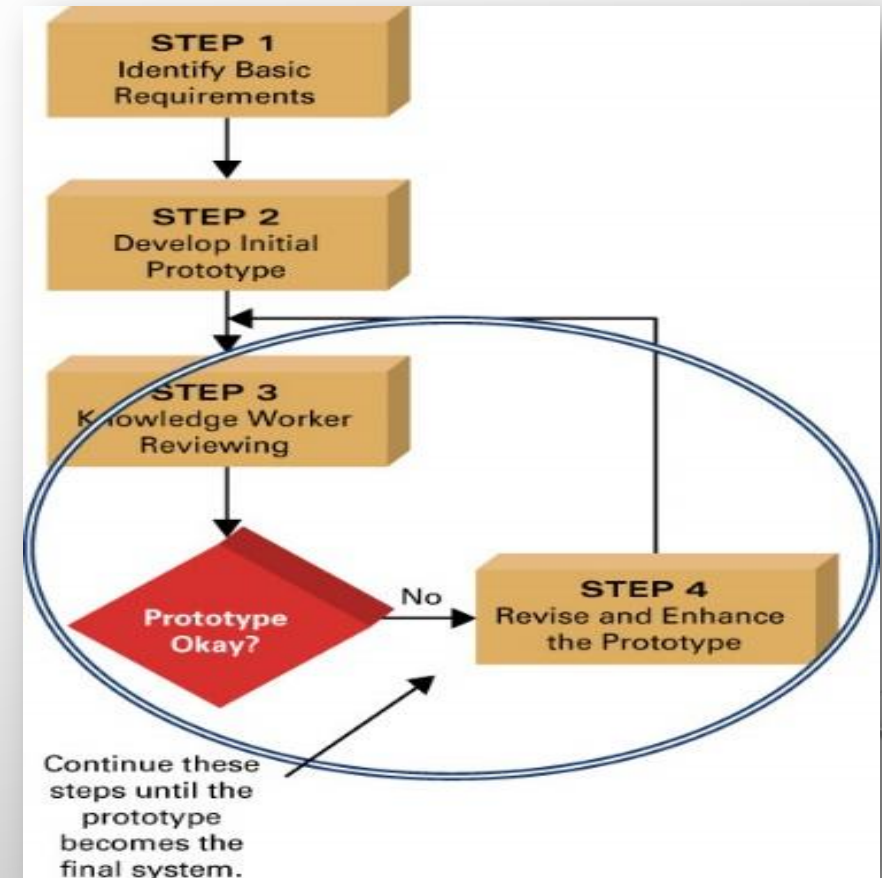


SYSTEMS DEVELOPMENT METHODOLOGIES



Advantages of Prototype Model

- ✓ Show the prototype to the client to have a clear understanding and complete 'feel' of the functionality developed in the software. It ensures a greater level of customer satisfaction and comfort.
- ✓ Identify the scope of the refinement and accordingly accommodate new changes in the given requirements.
- ✓ Significantly reduce the risk of failure using this method and identify the potential risks at an early stage and moderation steps can be taken quickly.
- ✓ The communication between the software development team and the client makes a very good and conducive environment during a project.
- ✓ It helps in requirement gathering and requirement analysis when there is a lack of required documents.

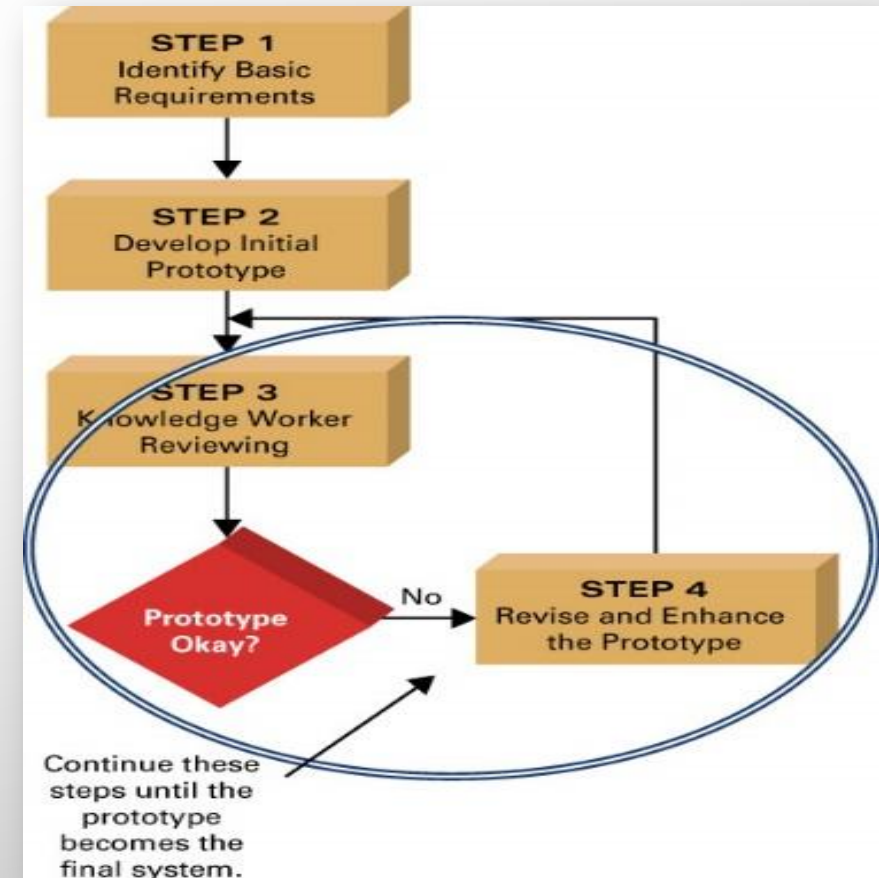




SYSTEMS DEVELOPMENT METHODOLOGIES

Disadvantages of Prototype Model

- ✓ Prototyping is usually done at the cost of the developer, so it should be done using minimal resources otherwise the organization's development cost stretches too much.
- ✓ Customers sometimes demand the actual product to be delivered soon after seeing an early prototype.
- ✓ The clients have too much involvement which is not always aligned with the software developer.
- ✓ It does not appreciate too many modifications in the project as it easily disturbs the existing workflow of the entire software development process.
- ✓ Customers may not be satisfied or interested in the product after seeing the initial prototype.

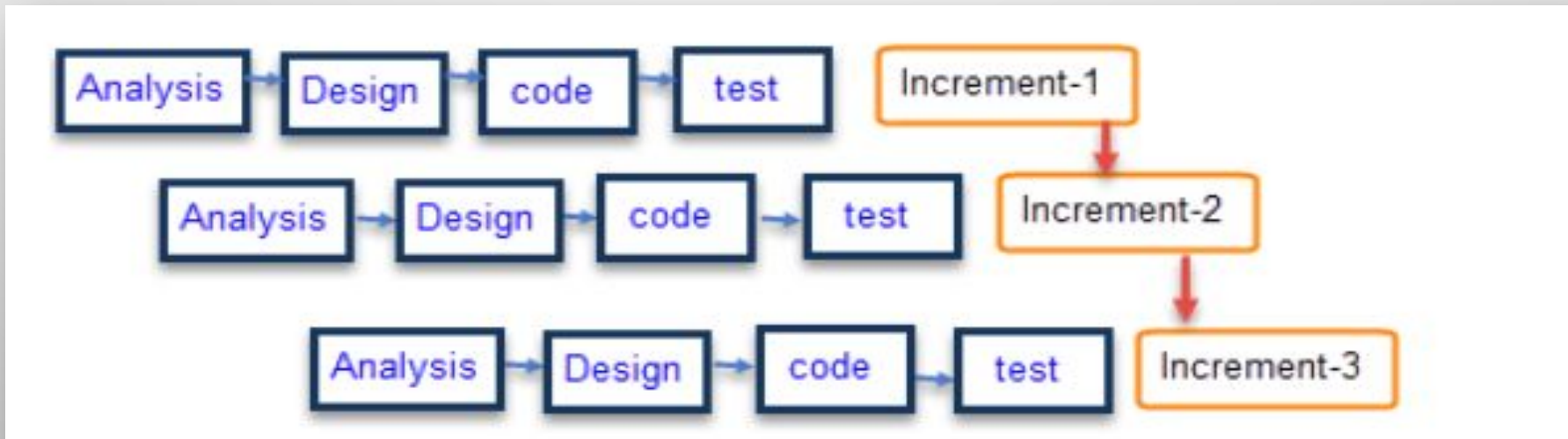




3. INCREMENTAL MODEL

A process of software development where requirements are broken down into multiple standalone modules of the software development cycle. Incremental development is done in steps from analysis design, implementation, testing/verification, maintenance.

Each iteration passes through the **requirements, design, coding and testing phases**. And each subsequent release of the system adds function to the previous release until all designed functionality has been implemented.





Advantages	Disadvantages
• The software will be generated quickly during the software life cycle	• It requires a good planning designing
• It is flexible and less expensive to change requirements and scope	• Problems might cause due to system architecture as such not all requirements collected up front for the entire software lifecycle
• Throughout the development stages changes can be done	• Each iteration phase is rigid and does not overlap each other
• This model is less costly compared to others	• Rectifying a problem in one unit requires correction in all the units and consumes a lot of time
• A customer can respond to each building	
• Errors are easy to be identified	

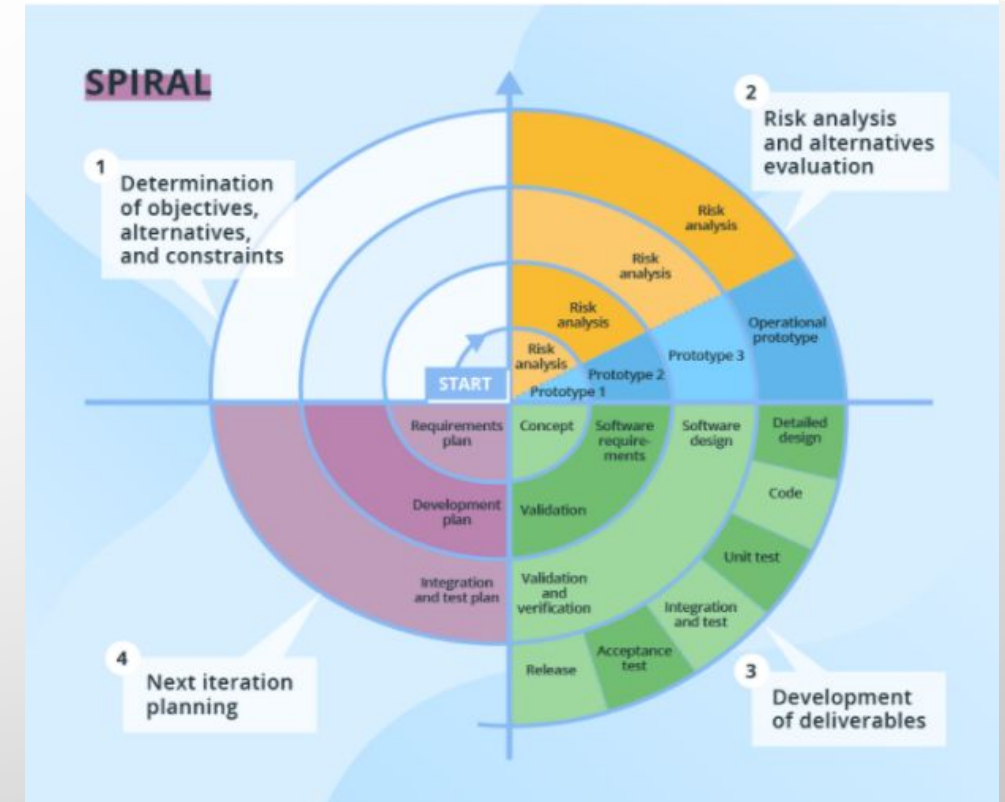
SYSTEMS DEVELOPMENT METHODOLOGIES



4. SPIRAL METHODOLOGY

A methodology for risk-based software development. It's a hybrid of the waterfall and iterative approaches. The Spiral Model assists in the adoption of software development aspects from several process models for software projects based on unique risk patterns, resulting in an efficient development process.

In the SDLC Spiral model, the development process begins with a small set of requirements and progresses through each development phase for those requirements. Until the application is ready for production, the software engineering team adds functionality for the new demand in ever-increasing spirals.



SYSTEMS DEVELOPMENT METHODOLOGIES



Spiral Model Phases

Planning

It includes estimating the cost, schedule and resources for the iteration. It also involves understanding the system requirements for continuous communication between the system analyst and the customer.

Risk Analysis

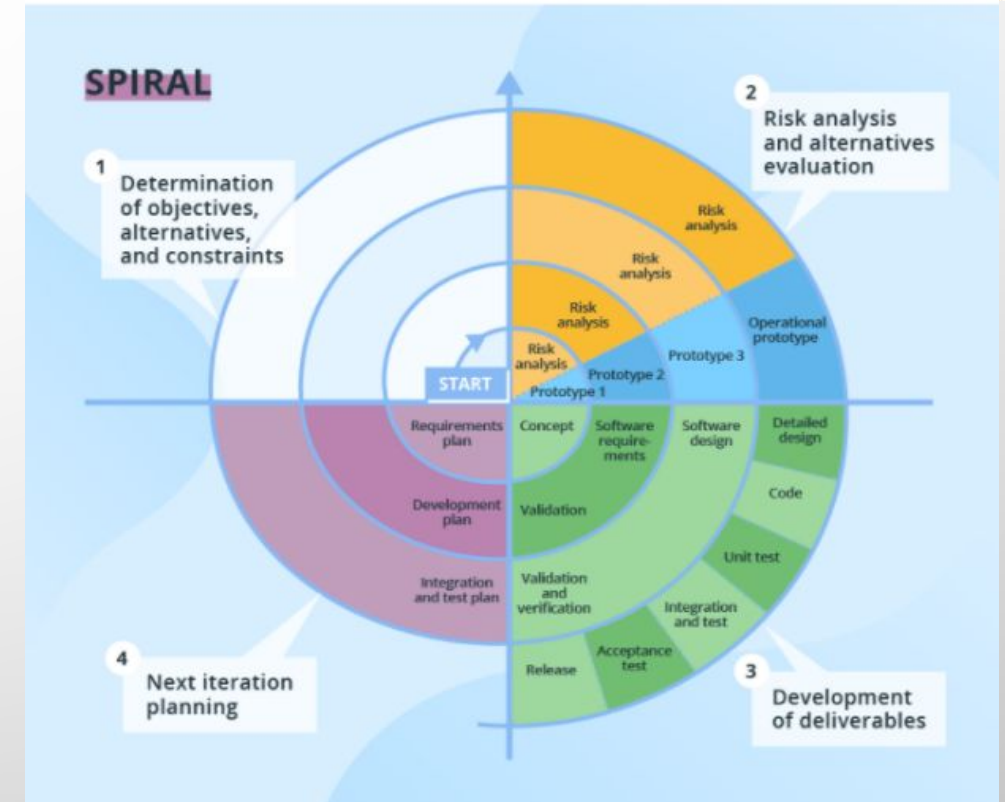
Identification of potential risk is done while risk mitigation strategy is planned and finalized.

Engineering

It includes testing, coding and deploying software at the customer site.

Evaluation

Evaluation of software by the customer. Also, includes identifying and monitoring risks such as schedule slippage and cost overrun.





Advantages	Disadvantages
• Additional functionality or changes can be done at a later stage	• Risk of not meeting the schedule or budget
• Cost estimation becomes easy as the prototype building is done in small fragments	• Spiral development works best for large projects only also demands risk assessment expertise
• Continuous or repeated development helps in risk management	• For its smooth operation spiral model protocol needs to be followed strictly
• Development is fast and features are added in a systematic way in Spiral development	• Documentation is more as it has intermediate phases
• There is always a space for customer feedback	• Spiral software development is not advisable for smaller project, it might cost them a lot

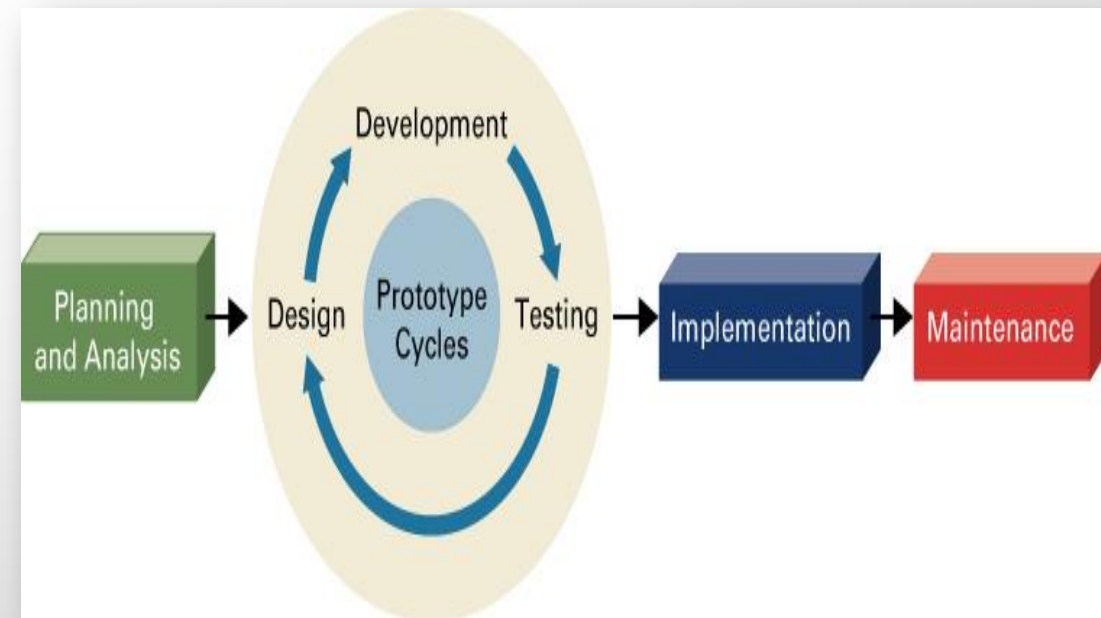


SYSTEMS DEVELOPMENT METHODOLOGIES

5. RAPID APPLICATION DEVELOPMENT (RAD)

Rapid application development (RAD) (also called rapid prototyping) - emphasizes extensive user involvement in the rapid and evolutionary construction of working prototypes of a system to accelerate the systems development process.

- Prototype - a smaller-scale, representation, or working model of the user's requirements or a proposed design for an information system,





SYSTEMS DEVELOPMENT METHODOLOGIES

Advantages of the RAD model

- ✓ Rapid Application development model helps to reduce the risk and required efforts on the part of the software developer.
- ✓ Additionally, this model also helps the clients to take quick reviews for the project.
- ✓ This methodology encourages customer feedback which always provides improvement scope for any software development project.
- ✓ As a result of prototyping in nature, there is a possibility of lesser defects.
- ✓ Each phase in RAD delivers the highest priority functionality to the client.



SYSTEMS DEVELOPMENT METHODOLOGIES



Disadvantages of the RAD model

- ✓ This model depends on the strong team and individual performances for clearly identifying the exact requirement of the business.
- ✓ It only works on systems that can be modularized and can be built using this methodology.
- ✓ This approach demands highly skilled developers and a designer's team which may not be possible for every organization.
- ✓ This method is not applicable for the developer to use in small budget projects as the cost of modeling and automated code generation is very high.
- ✓ Progress and problems accustomed are hard to track as such there is no documentation to demonstrate what has been done.

The Advantages & Disadvantages of the RAD Model



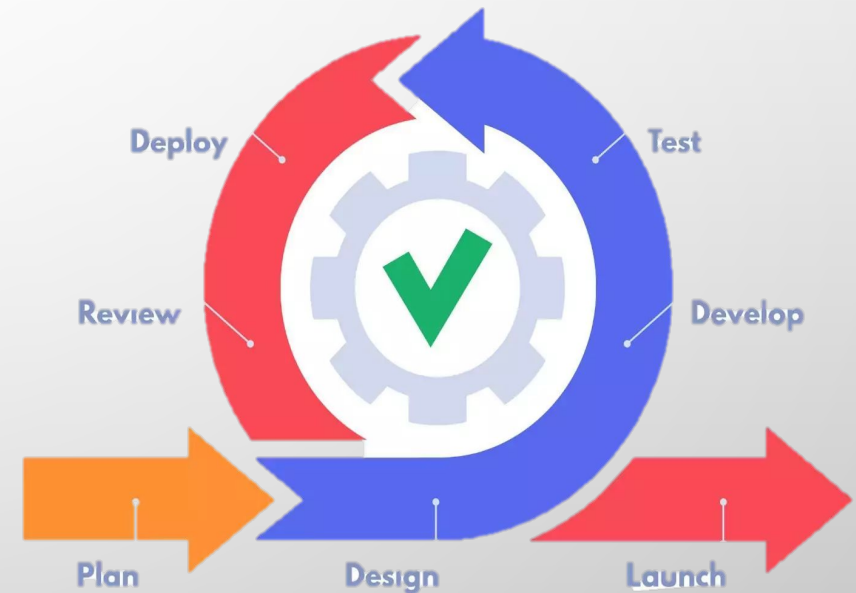


SYSTEMS DEVELOPMENT METHODOLOGIES

6. AGILE METHODOLOGY

A software development life cycle practice that encourages continuous iteration of development and testing. Unlike the Waterfall methodology, the Agile model in software testing allows for parallel development and testing.

One of the most straightforward and efficient methods for translating a business idea into technological solutions. Continuous planning, learning, improvement, team collaboration, evolutionary development, and early delivery are all terms used to describe agile software development methods. It promotes adaptability in the face of change.





SYSTEMS DEVELOPMENT METHODOLOGIES

FIVE STAGES OF AGILE METHODOLOGY

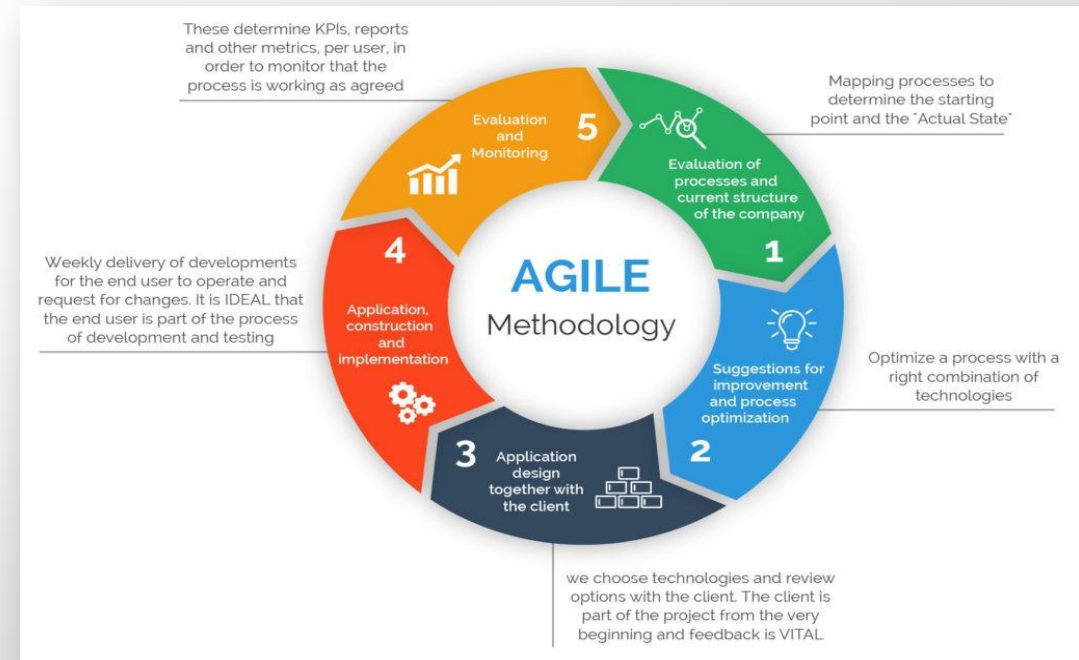
EXPLORATION

PLANNING

ITERATIONS TO THE FIRST RELEASE

PRODUCTIONIZING

MAINTENANCE





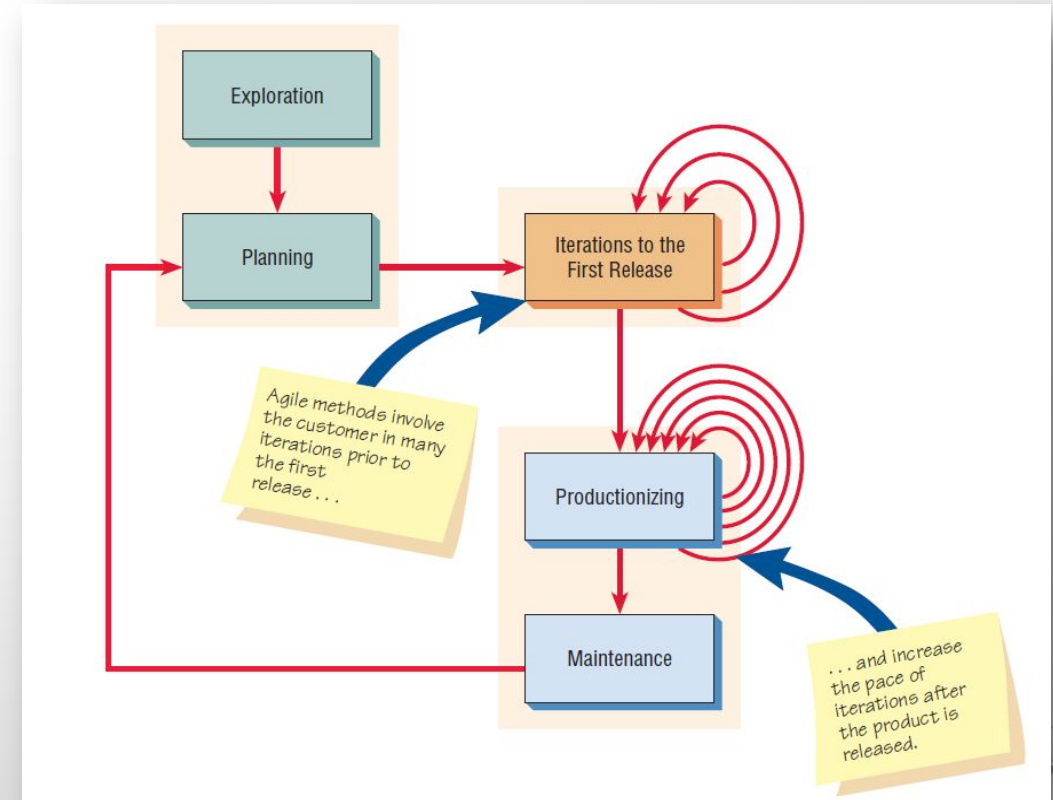
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METHODOLOGIES

EXPLORATION

During exploration, you will explore your environment, asserting your conviction that the problem can and should be approached with agile development, assemble the team, and assess team member skills.,





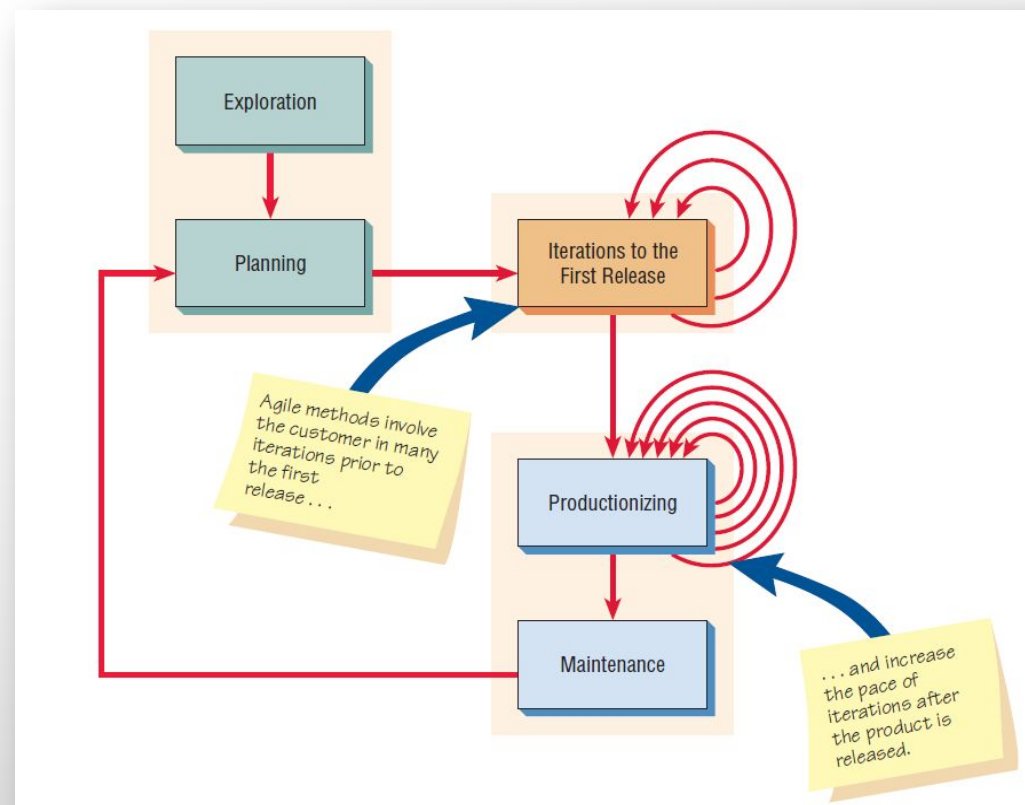
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DEVELOPMENT

METHODOLOGIES

PLANNING.

The next stage of the agile development process is called planning. In contrast to the first stage, planning may only take a few days to accomplish. In this stage you and your customers agree on a date anywhere from two months to half a year from the current date to deliver solutions to their most pressing business problems (you will be addressing the smallest, most valuable set of stories). If your exploration activities were sufficient, this stage



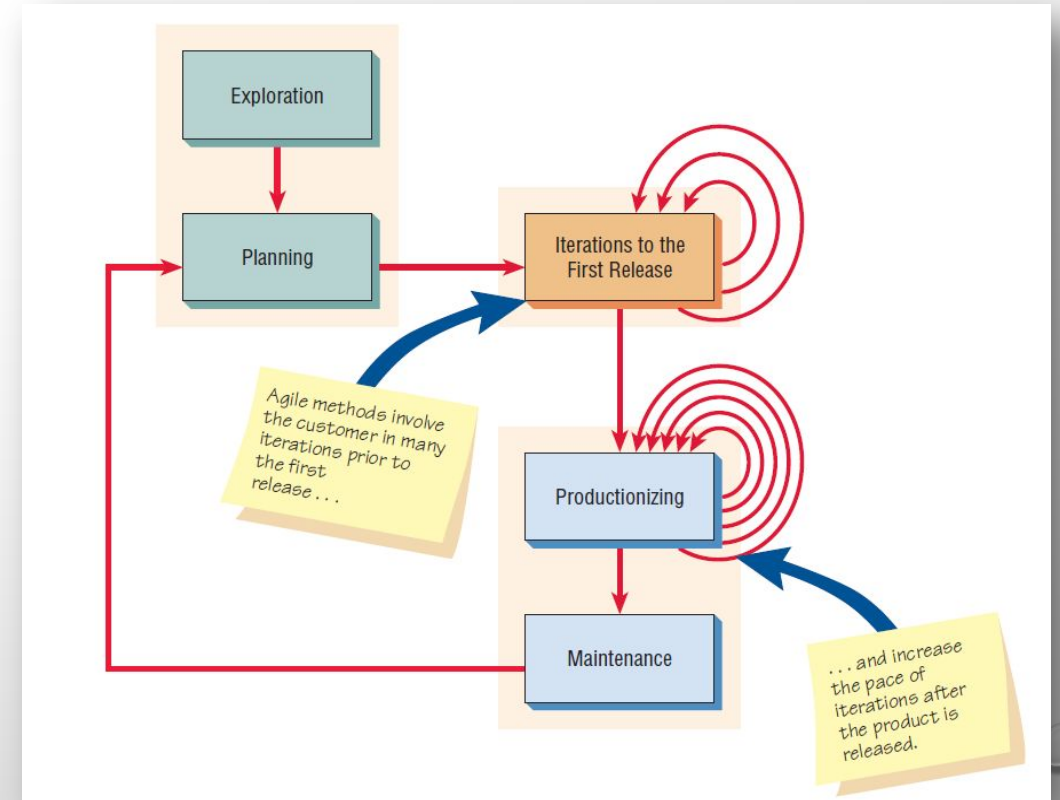
SYSTEMS

DEVELOPMENT

METHODOLOGIES

ITERATIONS TO THE FIRST RELEASE.

The third stage in the agile development process is composed of iterations to the first release. Typically, these are iterations (cycles of testing, feedback, and change) of about three weeks in duration. You will be pushing yourself to sketch out the entire architecture of the system, even though it is just in outline or skeletal form. One goal is to run customer-written functional tests at the end of each iteration. During the iterations stage you should also question whether the schedule needs to be altered or whether you are tackling too many stories



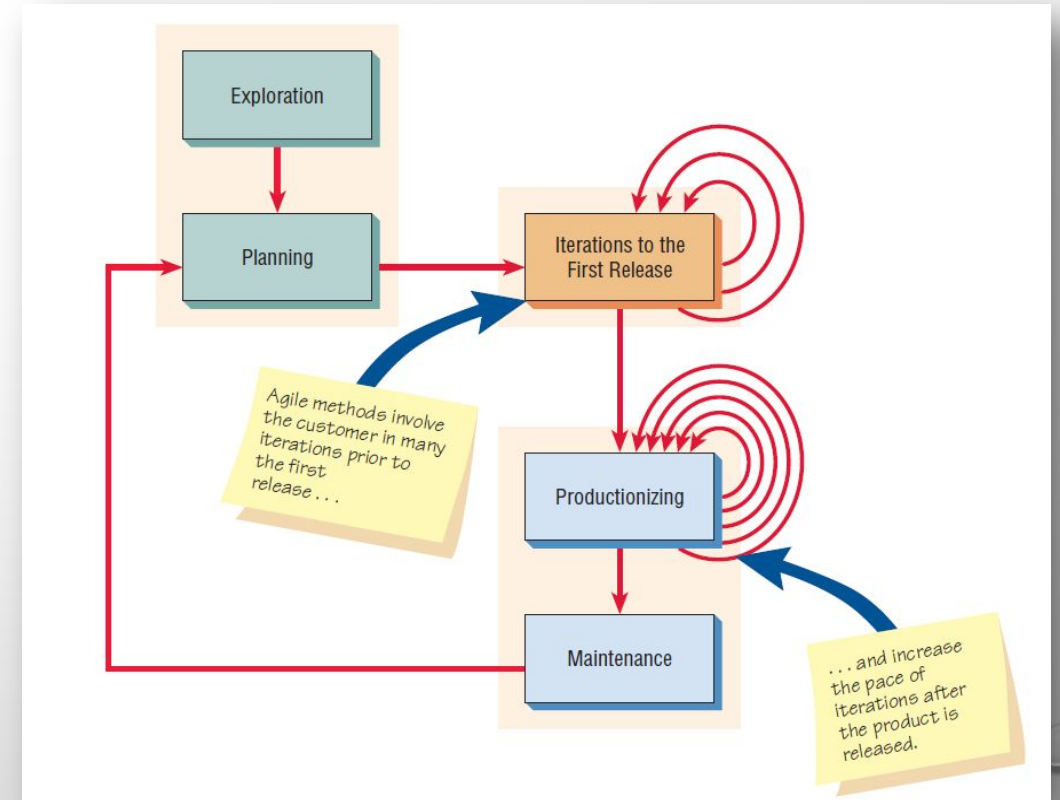
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PRODUCTIONIZING.

Several activities occur during this phase. In this phase the feedback cycle speeds up so that rather than receiving feedback for an iteration every three weeks, software revisions are being turned around in one week. You may institute daily briefings so everyone knows what everyone else is doing. The product is released in this phase, but may be improved by adding other features.



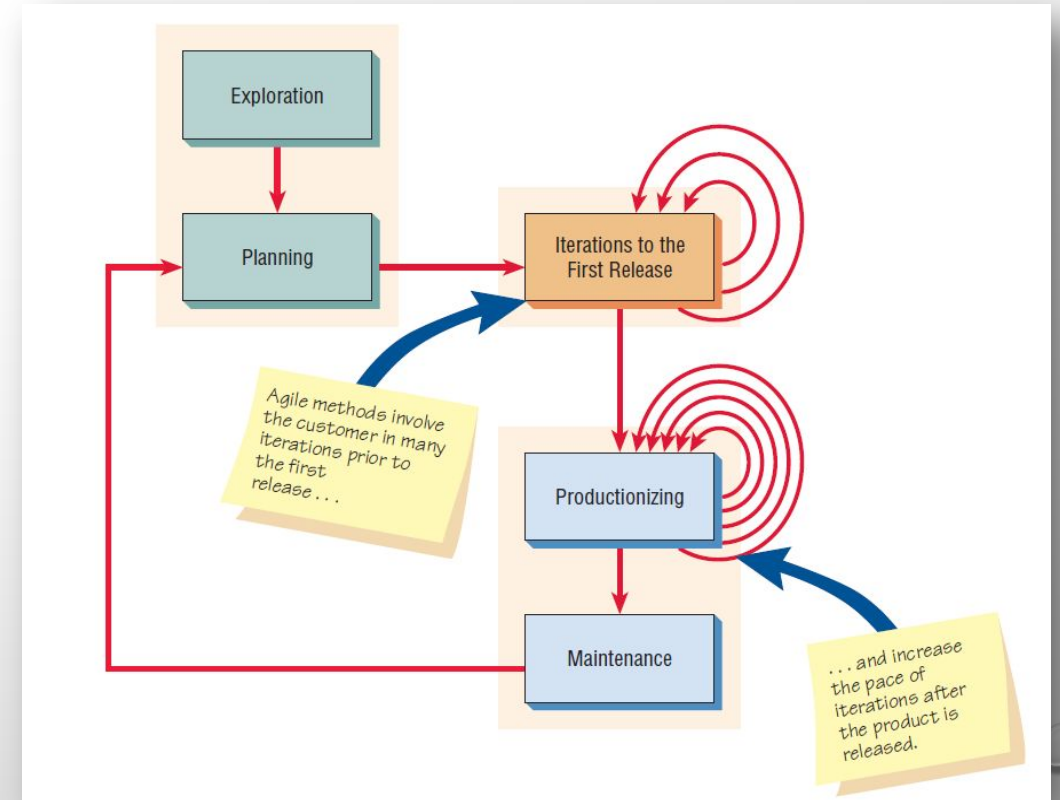
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MAINTENANCE.

Once the system has been released, it needs to be kept running smoothly. New features may be added, riskier customer suggestions may be considered, and team members may be rotated on or off the team.



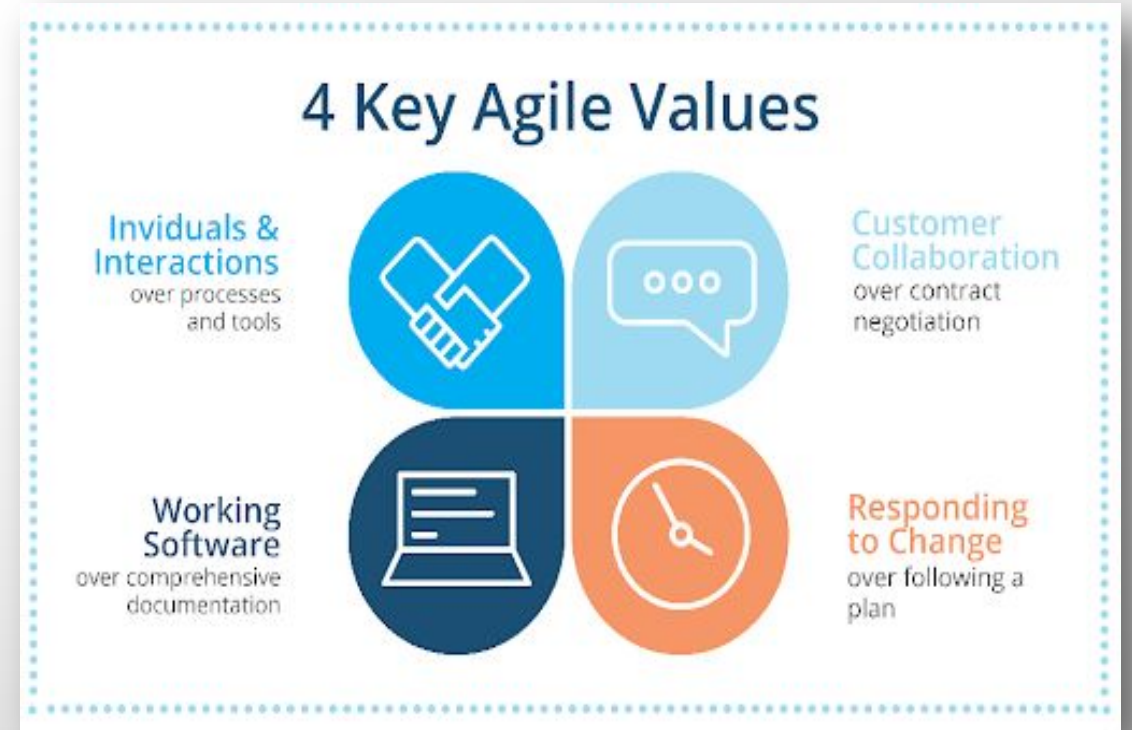
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Agile software development emphasizes four core values.

1. Individual and team interactions over processes and tools
2. Working software over comprehensive documentation
3. Customer collaboration over contract negotiation
4. Responding to change over following a plan



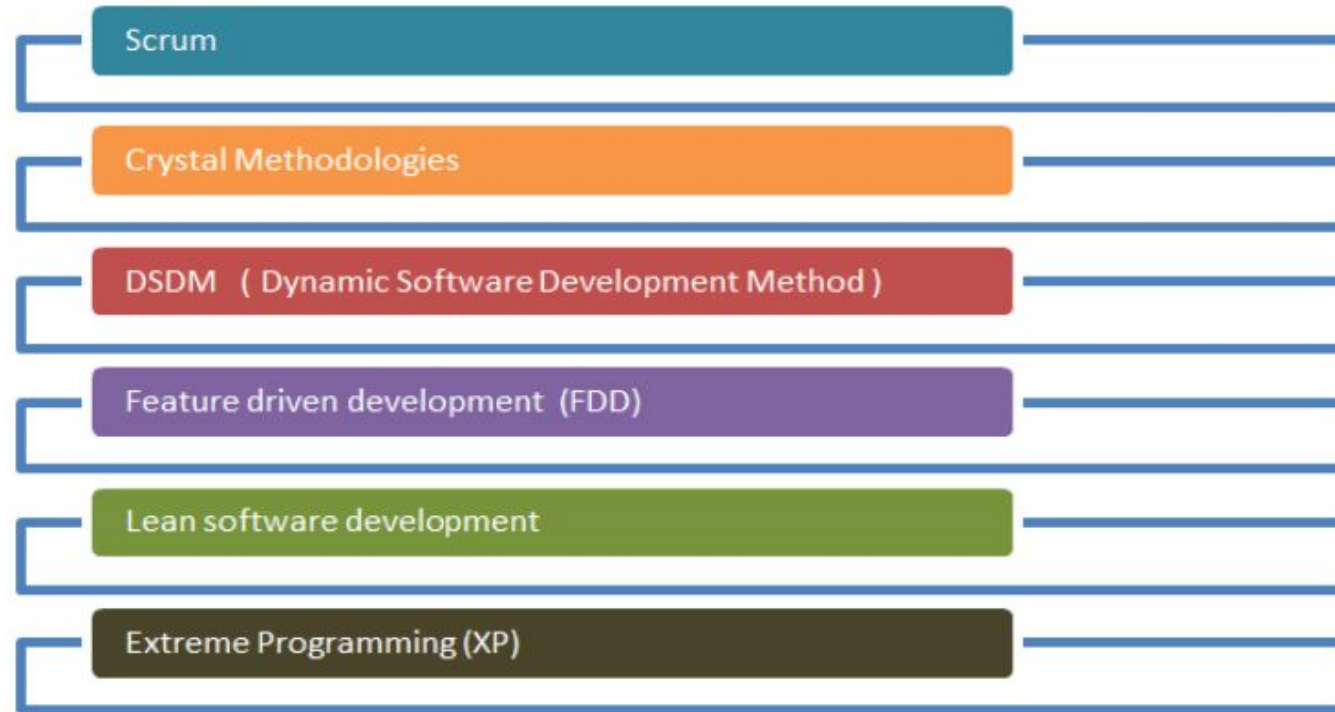
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Agile Process Model



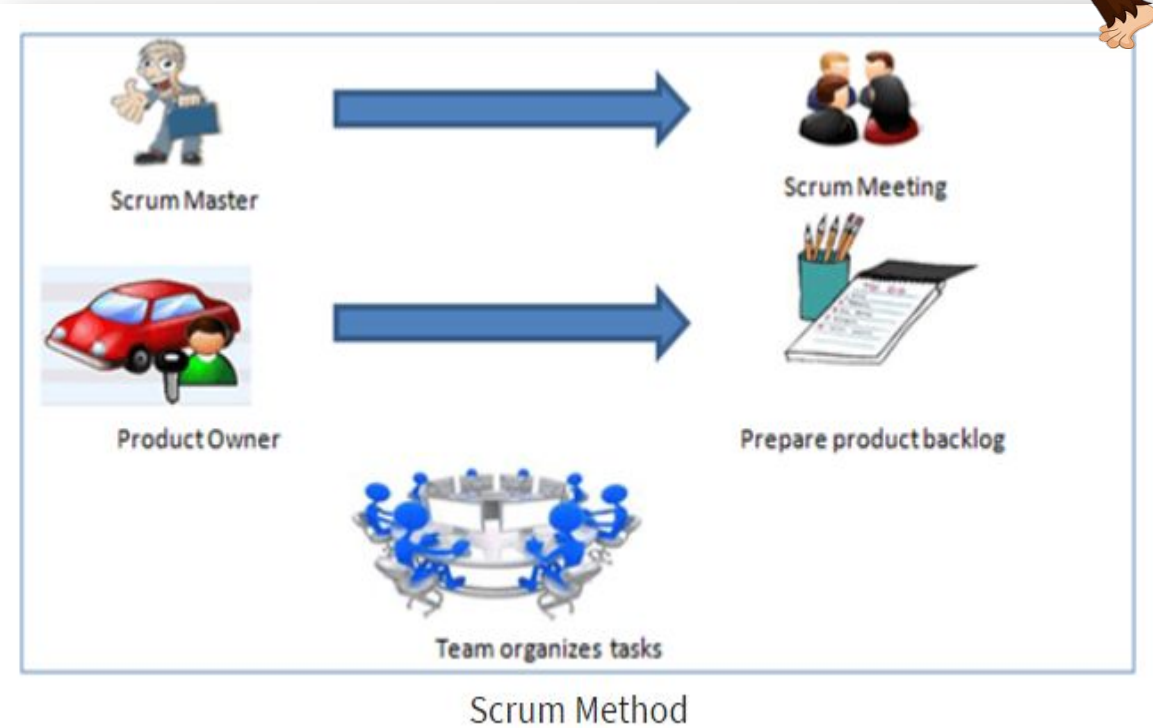
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Scrum

SCRUM is an agile development method which focuses on task management in a collaborative development setting. Scrum is a rugby term that refers to a group of people who work together to solve a problem. Scrum encourages working in small groups and believes in empowering the development team (say- 7 to 9 members). The following are the responsibilities of the three roles in Agile and Scrum:



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Scrum Master

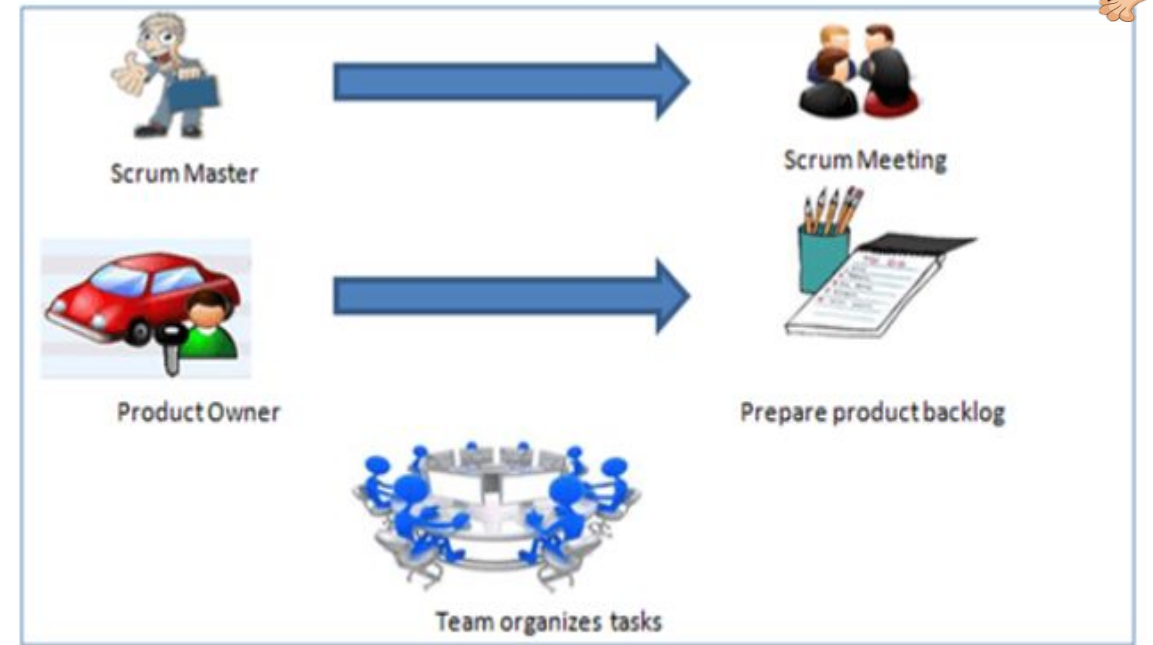
Master is responsible for setting up the team, sprint meeting and removes obstacles to progress.

Product owner

The Product Owner creates product backlog, prioritizes the backlog and is responsible for the delivery of the functionality at each iteration.

Scrum Team

Team manages its own work and organizes the work to complete the sprint or cycle.



Scrum Method

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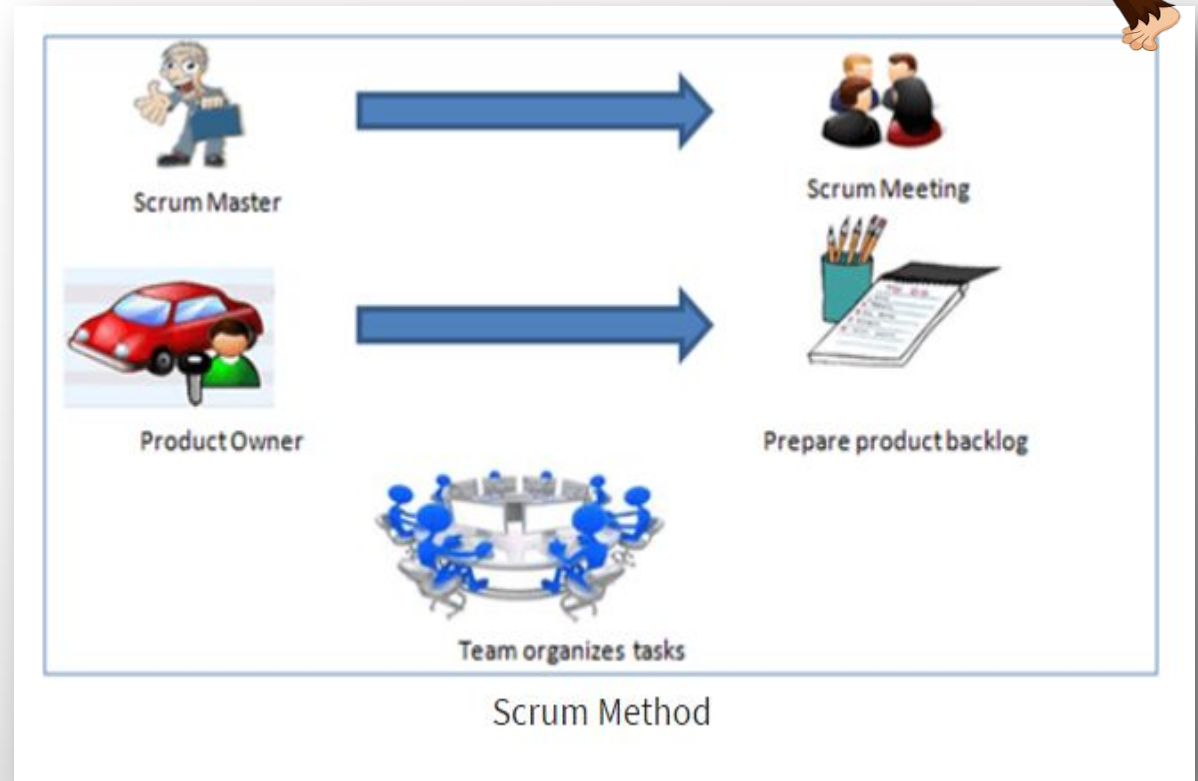
Product Backlog

This is a repository where requirements are tracked with details on the number of requirements(user stories) to be completed for each release. It should be maintained and prioritized by the Product Owner, and it should be distributed to the scrum team. Team can also request for a new requirement addition or modification or deletion.

Process flow of Scrum Methodologies:

Process flow of scrum testing is as follows:

- ✓ Each iteration of a scrum is known as Sprint.
- ✓ Product backlog is a list where all details are entered to get the end-product.
- ✓ During each Sprint, top user stories of Product backlog are selected and turned into Sprint backlog.
- ✓ Team works on the defined sprint backlog.
- ✓ Team checks for the daily work.
- ✓ At the end of the sprint, team delivers product functionality.



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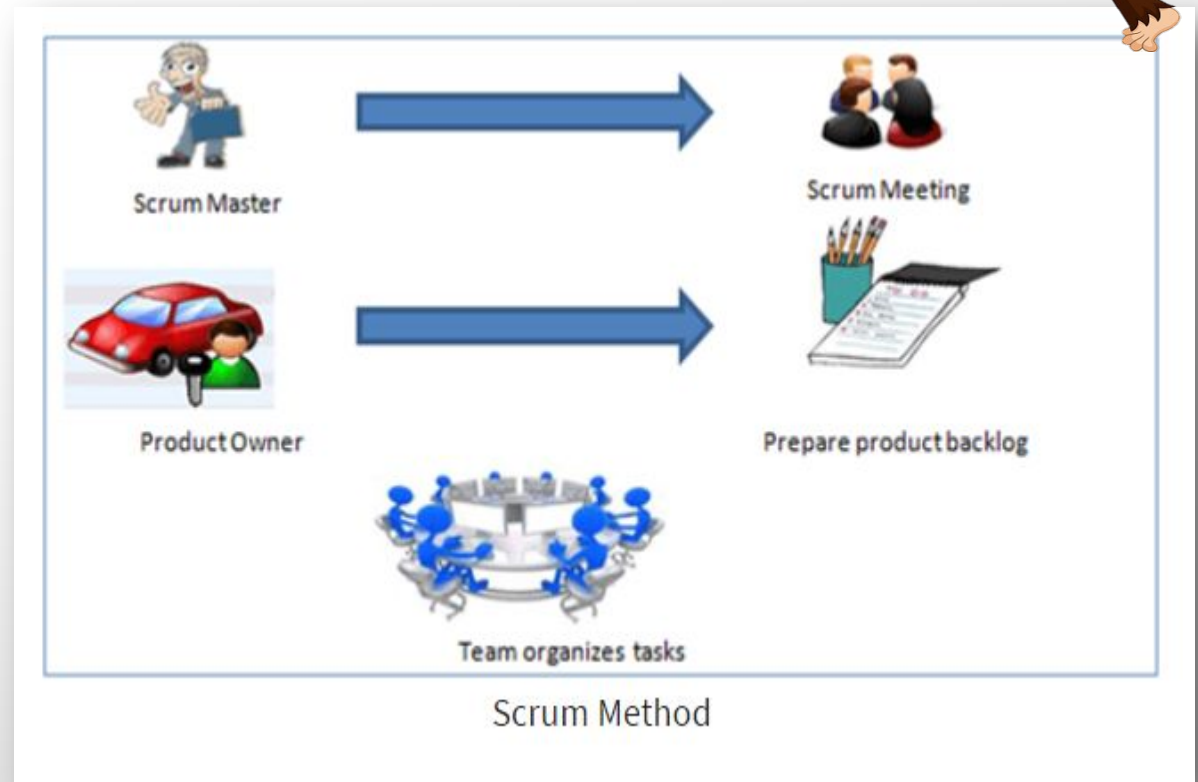
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Extreme Programming

When clients' needs or specifications are continually changing, or when they are unsure about the system's functionality, the Extreme Programming technique comes in handy. It encourages numerous "releases" of the product in short development cycles, which increases the system's efficiency and provides a checkpoint where any client requirements may be quickly incorporated. XP creates software with the client in mind.

Phases of eXtreme programming:

There are 6 phases available in Agile XP method, and those are explained as follows:



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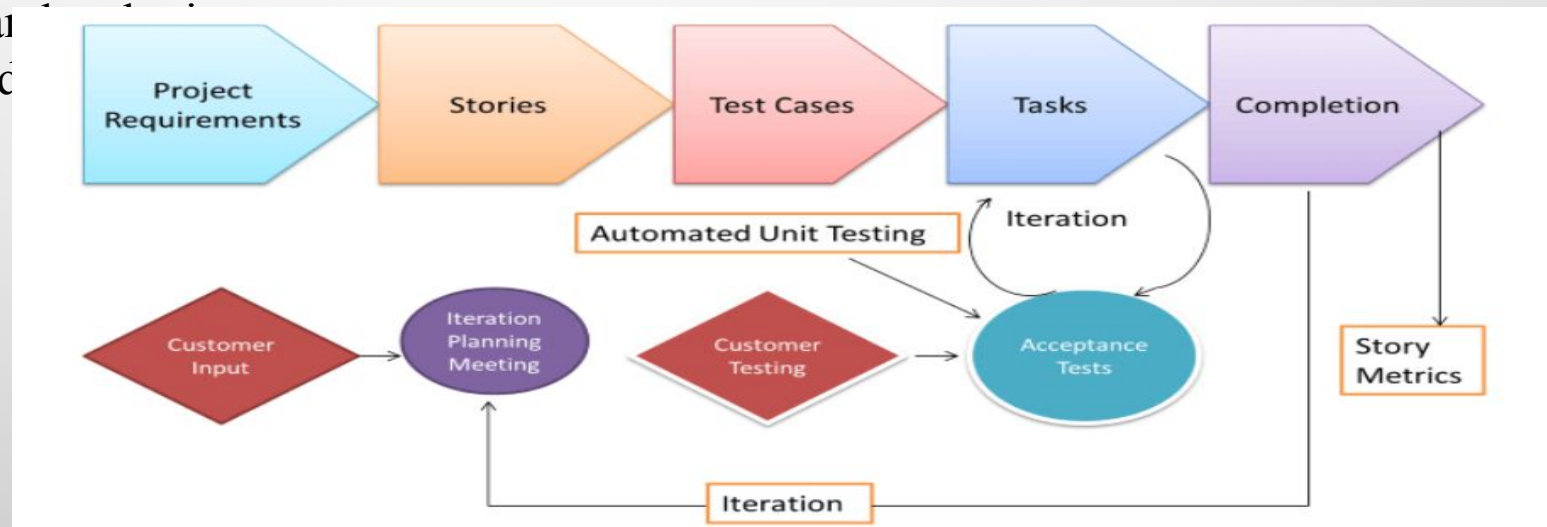
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Planning

Identification of stakeholders and sponsors
Infrastructure Requirements

Security related information and
Service Level Agreements and



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Analysis

Capturing of Stories in Parking lot

Prioritize stories in Parking lot

Scrubbing of stories for estimation

Define Iteration SPAN(Time)

Resource planning for both Development and QA teams

Design

Break down of tasks

Test Scenario preparation for each task

Regression Automation Framework

Execution

Coding

Unit Testing

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Execution of Manual test scenarios

Defect Report generation

Conversion of Manual to Automation regression test cases

Mid Iteration review

End of Iteration review

Wrapping

Small Releases

Regression Testing

Demos and reviews

Develop new stories based on the need

Process Improvements based on end of iteration review comments

Closure

Pilot Launch

Training

Production Launch

SLA Guarantee assurance

Review SOA strategy

Production Support



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Crystal Methodologies

Crystal Methodology is based on three concepts:

Chartering: Various activities involved in this phase are creating a development team, performing a preliminary feasibility analysis, developing an initial plan and fine-tuning the development methodology

Cyclic delivery: The main development phase consists of two or more delivery cycles, during which the

- Team updates and refines the release plan

- Implements a subset of the requirements through one or more program test integrate iterations

- Integrated product is delivered to real users

- Review of the project plan and adopted development methodology

Dynamic Software Development Method (DSDM)

DSDM is a Rapid Application Development (RAD) approach to software development and provides an agile project delivery framework.

The important aspect of DSDM is that the users are required to be involved actively, and the teams are given the power to make decisions. Frequent delivery of product becomes the active focus with DSDM. The techniques used in DSDM are:

- ✓ Time Boxing
- ✓ MoSCoW Rules
- ✓ Prototyping

Wrap Up: The activities performed in this phase are deployment into the user environment, post- deployment

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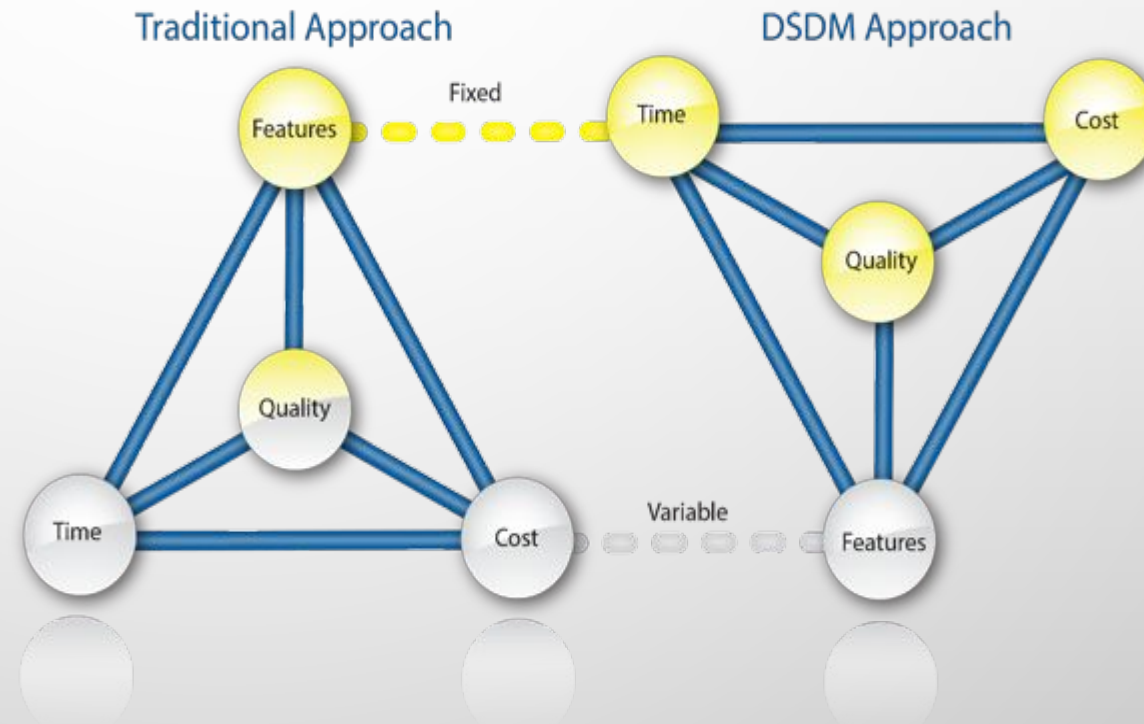
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The DSDM project consists of 7 phases

1. Pre-project
2. Feasibility Study
3. Business Study
4. Functional Model Iteration
5. Design and build Iteration
6. Implementation
7. Post-project



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Feature Driven Development (FDD)

This method is focused around "designing & building" features. Unlike other Agile methods in software engineering, FDD describes very specific and short phases of work that has to be accomplished separately per feature. It includes domain walkthrough, design inspection, promote to build, code inspection and design. FDD develops product keeping following things in the target.

1. Domain object Modeling
2. Development by feature
3. Component/ Class Ownership
4. Feature Teams
5. Inspections
6. Configuration Management
7. Regular Builds
8. Visibility of progress and results



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Lean Software Development

Lean software development method is based on the principle "Just in time production". It aims at increasing the speed of software development and decreasing cost. Lean development can be summarized in seven steps.

1. Eliminating Waste
2. Amplifying learning
3. Defer commitment (deciding as late as possible)
4. Early delivery
5. Empowering the team
6. Building Integrity
7. Optimize the whole



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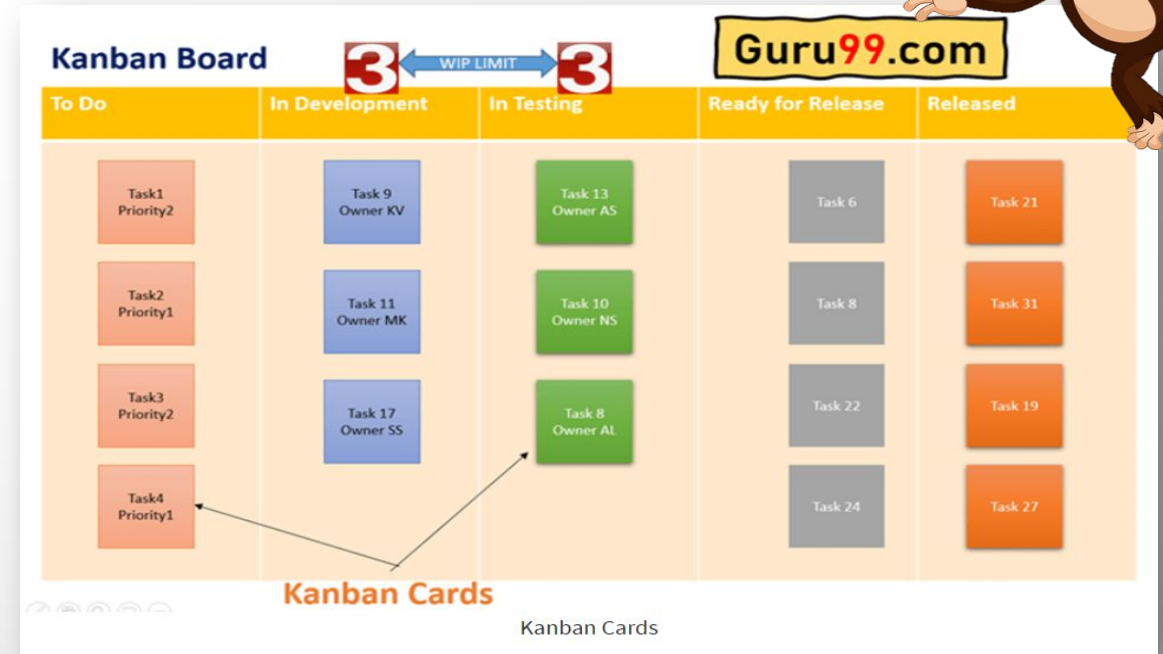
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Kanban

Kanban originally emerged from Japanese word that means, a card containing all the information needed to be done on the product at each stage along its path to completion. This framework or method is quite adopted in software testing methods especially in Agile concepts.

It gives a clear manner of viewing a team's tasks and work capability. Physical and digital boards are primarily used to allow team members to visualize the present condition of the project they are working on. The Kanban system recommends visualization of work. It suggests the use of the physical and the digital board.



The Kanban cards are essential pieces on the Kanban board as it represents the work that the team is working on. These cards will have:

- ✓ Priority
- ✓ Owner
- ✓ Type
- ✓ Due date